

GRADUATION PROJECT

Industry Internship: January - May 2026

Volume: 1/1

INTERNSHIP: **OIMake**

STUDENT: **D. SRI CHARAN REDDY**

ENROLLMENT ID: **22WU0301104**

GUIDE: **Proff. BHUSHAN SONAWANE**

PROGRAMME: **BACHELORS OF DESIGN (B.DES)**

2026

INDUSTRIAL DESIGN

ORIGINALITY STATEMENT

I, Sri Charan Reddy, son of Srinivas Reddy, a bonafide graduating student with Reg. No. 22WU0301104 of School of Arts and Design (SOAD), Woxsen University, certify that the work in this Graduation Design, LED Light Bag, submitted on 20/05/2026, for the partial fulfilment of the UG degree in B.Des. Industrial Design of Batch 2022- 2026, is my original work and has not been submitted previously for any degree or Degree Project.

This report, authored by me under the guidance of my project mentors, acknowledges all assistance received. The intellectual content is my own, with due credit given for all sources used. This project has not been and will not be submitted for assessment in any other course. None of the concepts are borrowed or copied without acknowledgment, and all assistance in design, conception, style, presentation, and linguistic expression is duly credited.

Certified by:

Name: Duggyampudi Sri Charan Reddy

Signature:

Date:

Place:

Verified by:

Faculty Mentor: Bhushan Sonwane

Signature:

Date:

Place:

CERTIFICATE

This is to certify that the Design Degree Project titled “_DIY STEAM Toys_” has been completed by D Sri Charan Reddy as part of the curriculum requirement for the Bachelor of Design degree. The final work was evaluated and has met the required academic and professional standards set by the School of Arts and Design (SOAD), Woxsen University.

Project Details: I am developing **STEAM DIY toy kits** under the brand **OiMake** — hands-on electronics projects for children aged 8 and above, where kids build, paint, and light up real objects they keep and use daily.

Project Title: ___DIY STEAM Toys___

Project Description: __**STEAM DIY Toys** Hands-on kits where children build, create, and learn through making.

LED LightBag A DIY kit — paint a bag, build the lights, wear it.

Light Up Fridge Magnet A DIY kit — stack the layers, insert electronics, press to glow.

Duration: 06 January 2026 to 30 June 2026

Faculty Advisor: Bhushan Sonawane

School of Arts and Design (SOAD) Woxsen University

Date: __/__/____

Dr. Adity Saxena

Head of Design Department

School of Arts and Design (SOAD) Woxsen

University

Seal of the Institution

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Student Name: Duggyampudi Sri Charan Reddy

Signature:

Date:

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Bachelor's Program in Design,

Product Design, 2022-2026

Woxsen University, Hyderabad, Telangana, India

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May 2026

JURY EVALUATION SHEET

The Evaluation Jury feedback of the Degree Project, presented in the School of Arts and Design (SOAD), Woxsen University by (Name of the student) on (Project Title).

Department: _____ Date of Evaluation: _____

Jury Members:

Name:

Designation:

Signature:

Comments:

Name:

Designation:

Signature:

Comments:

Name:

Designation:

Signature:

Comments:

JOINING REPORT

This is to certify that Mr._D Sri Charan Reddy____
Reg.No __22WU0301104_ of Eighth Semester B.Des
(Industrial Design), Woxsen University School of Arts and
Design (SOAD), has joined as an “Intern” at our
office/firm/organization on _OIMake_. and he will continue his
training till _30th June 2026__Our office works for 6 days a
week from _9.30__AM to _6.30_PM.



Signature and seal of the Head
of the Organization.

27th Dec. 2025

Industrial Design Internship

Dear Sri Charan,

We are pleased to offer you the position of Industrial Design intern at
Makercubs Technologies Pvt. Ltd.

1. Your internship duration will be from 6th Jan. 2026 to 30th Jun 2026.
2. Your remuneration will be INR 12,000 per month.
3. Office working hours are 9.30AM to 6.30PM, 6 days/week.
4. You will be working on Design and development of DIY tech toys.
Apart from working on your graduation project, You will be involved in
other projects, day to day work of the organization.
5. As an organization policy, you will be required to sign a
Non-disclosure agreement for your graduation project.

We welcome you to be a part of a team of passionate makers and look
forward to having a great working relationship with you.

Thank you.



Kshiteesh Kulkarni
Chief Maker @ MakerVerse

Office address:
Makercubs Technologies Pvt. Ltd.
2nd floor, Star commercial center
Opposite Gitanjali Vedika school,
Puppalguda, Hyderabad
Telangana 500089

MONTHLY PROGRESS REPORT

Name	Sri Charan Reddy
Roll No	22WU0301104
Institute/University	Woxsen University
Report for Month	January 2026
Number of working days	26 Days(From 06/01/2026 to 31/01/2026)
Number of days attended	24 Working Days (2 Leave Days)

Name of the Projects and Work Details

1. LEGO SmartPlay Study

- Understanding LEGO SmartPlay working and interaction system
- Studying product mechanism and user engagement

2. DIY Drawstring Bag for Kids

- Secondary and market research
- Anthropometric study for children aged 8–14
- Bag dimension development
- Graphic and illustration exploration
- Prototype development
- Painting and material experimentation
- LED and lighting integration concepts
- Pocket placement and construction development

3. Sakura Lamp

- Assembly study and understanding
- Step-by-step assembly documentation
- Picture manual development
- Manual testing and corrections
- Assembly process recapturing and refinement

4. Lighting Pattern Development

- Pattern creation for illuminated paintable objects
- EVA foam prototype development
- Material stiffness and structure testing

Declaration: All the above information provided by the student is true as per records.

Signature of the
Candidate



Signature of the Project
Head Designer

MONTHLY PROGRESS REPORT

Name	Sri Charan Reddy
Roll No	22WU0301104
Institute/University	Woxsen University
Report for Month	February 2026
Number of working days	24 Days(From 06/01/2026 to 31/01/2026)
Number of days attended	23 Working Days (1 Leave Days)

Name of the Projects and Work Details

1. DIY Drawstring Bag for Kids

- Illustration development and refinement
- Material and coloring experiments
- Acrylic, marker, and washable pen testing
- Cloth and visibility testing
- Bag dimension refinement
- EVA foam and pocket development
- Photo manual creation
- BOM preparation
- User testing preparation

2. Fridge Magnet

- DXF file creation
- Prototype development
- Mandala pattern design
- Laser cutting tests
- PCB arrangement and LED setup
- Light diffusion experimentation
- Push-lock button interaction development
- MDF and LGP structure testing
- Adjustable jig development

3. Workshops and User Testing

- School workshop preparation
- Conducting workshops with children
- Product feedback collection
- Illustration and usability improvements

Declaration: All the above information provided by the student is true as per records.

Signature of the
Candidate



Signature of the Project
Head Designer

MONTHLY PROGRESS REPORT

Name	Sri Charan Reddy
Roll No	22WU0301104
Institute/University	Woxsen University
Report for Month	March 2026
Number of working days	26 Days(From 06/01/2026 to 31/01/2026)
Number of days attended	22 Working Days (2 Leave Days)

Name of the Projects and Work Details

1. DIY Drawstring Bag for Kids

- Illustration refinement based on feedback
- Girls illustration development
- Vendor coordination for stitching
- Dimension and construction finalization
- Tracing method development
- User testing with children
- Product usability improvements

2. Instruction Manual Development

- Manual layout preparation
- Illustration-based instruction creation
- Product photography documentation
- 6-LED bag manual development
- 5-LED bag manual planning and preparation

3. Fridge Magnet

- Illustration assembly development
- Screw and rubber-band assembly experimentation
- Production file cleaning and organization

4. Workshops

- Conducting DIY bag workshops
- Product demonstration and feedback analysis

Declaration: All the above information provided by the student is true as per records.

Signature of the
Candidate



Signature of the Project
Head Designer

MONTHLY PROGRESS REPORT

Name	Sri Charan Reddy
Roll No	22WU0301104
Institute/University	Woxsen University
Report for Month	April 2026
Number of working days	26 Days(From 06/01/2026 to 31/01/2026)
Number of days attended	24 Working Days (2 Leave Days)

Name of the Projects and Work Details

1. Fridge Magnet

- Panel file preparation
- Illustrator file cleaning
- Manual development
- BOM preparation
- Final assembly documentation

2. DIY Drawstring Bag for Kids

- Final manual preparation
- Documentation refinement
- Production file organization

3. Appuworld

- Product-related development tasks
- Sample preparation

4. LDR Sensor Based Projects

- LDR project research
- Interactive educational product exploration
- Fortune wheel concept development
- Timer and gramophone project ideation
- Buzzer integration planning

5. Workshop Kits

- Football kit preparation
- Workshop material arrangement

Declaration: All the above information provided by the student is true as per records.

Signature of the
Candidate



Signature of the Project
Head Designer

MONTHLY PROGRESS REPORT

Name	Sri Charan Reddy
Roll No	22WU0301104
Institute/University	Woxsen University
Report for Month	April 2026
Number of working days	26 Days(From 06/01/2026 to 31/01/2026)
Number of days attended	24 Working Days (2 Leave Days)

Name of the Projects and Work Details

1. Fridge Magnet

- Panel file preparation
- Illustrator file cleaning
- Manual development
- BOM preparation
- Final assembly documentation

2. DIY Drawstring Bag for Kids

- Final manual preparation
- Documentation refinement
- Production file organization

3. Appuworld

- Product-related development tasks
- Sample preparation

4. LDR Sensor Based Projects

- LDR project research
- Interactive educational product exploration
- Fortune wheel concept development
- Timer and gramophone project ideation
- Buzzer integration planning

5. Workshop Kits

- Football kit preparation
- Workshop material arrangement

Declaration: All the above information provided by the student is true as per records.

Signature of the
Candidate



Signature of the Project
Head Designer

PRACTICAL TRAINING (2026-26)

MONTHLY LOG SHEET Januaru-May, 2026

Name of the Trainee: D Sri Charan Reddy

Name of the Firm: OIMake

Date	Day	Project Details	Nature of Work
06/01/26	Tuesday	LEGO SmartPlay & DIY Bag Research	Understood LEGO SmartPlay working at CES, started secondary and market research on DIY bags for kids
07/01/26	Wednesday	DIY Bag Research	Continued secondary and market research on DIY bags
08/01/26	Thursday	Sakura Lamp Assembly	Reviewed research, started Sakura lamp assembly study, documented assembly process step-by-step
09/01/26	Friday	Sakura Lamp Manual	Created picture manual, implemented feedback changes, recaptured assembly process, completed picture manual
10/01/26	Saturday	Sakura Lamp Testing & DIY Bag Materials	Conducted assembly testing with intern using manual, noted improvements, sourced materials for DIY bag experiments
12/01/26	Monday	Manual Finalization & Bag Experimentation	Finalized picture manual, conducted trial-and-error for bag painting process and manufacturing research
13/01/26	Tuesday	DIY Bag Prototype	Developed prototype and explored project direction
14/01/26	Wednesday	Anthropometric Research	Researched anthropometric measurements for children aged 8–14, finalized bag dimensions and graphics
15/01/26	Thursday	Leave	Leave
17/01/26	Saturday	Leave	Leave
19/01/26	Monday	Lighting Pattern Development	Developed lighting patterns for paintable objects, created prototype, identified need for thicker foam
20/01/26	Tuesday	LED Integration Discussion	Discussed bag development, LED mounting concepts, PCB spacers, and 6-LED graphic patterns

21/01/26	Wednesday	Testing & Graphic Design	Conducted battery drain tests, tested laser cutting on muslin, designed graphics and patterns
22/01/26	Thursday	Graphic Development	Continued graphics and pattern development
23/01/26	Friday	Prototype Finalization	Completed graphics for prototype, finalized dimensions, modified pocket placement, initiated sample prototype
24/01/26	Saturday	Lighting Setup Iteration	Reworked lighting setup for the bag
27/01/26	Tuesday	Illustration Development	Received completed bag prototype and started illustration work
28/01/26	Wednesday	Market Observation & Illustration Study	Continued illustrations, conducted observational study at malls and toy stores
29/01/26	Thursday	Illustration Development	Developed illustrations
30/01/26	Friday	Prototype Painting & Final Dimensions	Created clean painted prototype, finalized dimensions and EVA foam sizing
31/01/26	Saturday	Illustration Finalization	Worked on finalizing illustrations
02/02/26	Monday	Illustration & Paint Testing	Continued illustrations and tested paint samples
03/02/26	Tuesday	Illustration Work	Continued extensive illustration work
04/02/26	Wednesday	Material Testing	Tested designs on muslin cloth, revised dimensions, sent materials
05/02/26	Thursday	Illustration Revision	Made illustration modifications
06/02/26	Friday	Coloring Technique Experiments	Tested acrylics, fabric markers, sketches, washable pens, and different white fabrics
07/02/26	Saturday	General Work	Minor work / updates

09/02/26	Monday	Fridge Magnet Project Start	Rechecked bag dimensions and coloring process, started fridge magnet project
10/02/26	Tuesday	Fridge Magnet Prototype	Created DXF file and fridge magnet prototype
11/02/26	Wednesday	Leave	Leave
12/02/26	Thursday	PCB Arrangement & Pattern Search	Rearranged PCB layout and explored fridge magnet patterns
13/02/26	Friday	Mandala Pattern Development	Designed mandala patterns for fridge magnet
14/02/26	Saturday	Laser Testing	Tested laser-cut patterns and identified file corrections
16/02/26	Monday	Light Dispersion Development	Updated files and experimented with light dispersion techniques
17/02/26	Tuesday	Lighting Setup & Manual Start	Continued light diffusion experiments and began DIY bag photo manual
18/02/26	Wednesday	Photo Manual & Jig Development	Completed photo manual and created adjustable jig for light diffusion testing
19/02/26	Thursday	Fridge Magnet Iteration	Iterated fridge magnet lighting setup
20/02/26	Friday	BOM & Material Testing	Prepared BOM for drawstring bag and tested non-woven material
21/02/26	Saturday	Material & Rendering Experiments	Tested lighting, diffusing materials, LGP casing, and installed KeyShot
23/02/26	Monday	Mandala & PCB Setup	Created mandala combinations and finalized LED PCB setup
24/02/26	Tuesday	Interaction Development	Developed push-lock button interaction for fridge magnet
25/02/26	Wednesday	PCB & MDF Setup	Finalized PCB and MDF arrangement, aligned DIY bag illustrations

26/02/26	Thursday	EVA Pocket File & Simulation	Worked on drawstring bag BOM, created EVA pocket file, conducted simulation
27/02/26	Friday	Kids Workshop Preparation	Conducted workshop with school kids and prepared for next workshop
28/02/26	Saturday	User Testing & Feedback	Completed prototypes, conducted workshop, gathered feedback and improvements
02/03/26	Monday	Illustration Revision	Updated illustrations based on feedback and searched for tracing alternatives
03/03/26	Tuesday	Illustration Development	Continued illustration work, especially girls' illustrations
05/03/26 – 09/03/26	Thursday–Monday	Illustration Finalization & Vendor Coordination	Continued illustration work and coordinated bag stitching order with vendor
10/03/26	Tuesday	Product Finalization	Finalized illustrations, dimensions, construction, and tracing
11/03/26	Wednesday	Manual Preparation	Reviewed finalized work and arranged materials for final manual
12/03/26	Thursday	Manual Illustration Work	Developed manual visuals and completed illustrations
13/03/26 – 14/03/26	Friday–Saturday	Manual Development	Continued manual preparation and image setup
16/03/26	Monday	Leave	Leave
17/03/26 – 18/03/26	Tuesday–Wednesd ay	Manual Work	Continued manual development
20/03/26	Friday	Workshop & Product Photography	Conducted workshop, updated product details, completed product photography
21/03/26 – 23/03/26	Saturday–Monday	Leave	Leave

24/03/26 – 26/03/26	Tuesday–Thursday	Drawstring Bag Manual	Finalized 6-LED bag manual and planned 5-LED manual
27/03/26 – 01/04/26	Friday–Wednesday	Fridge Magnet Assembly Development	Worked on illustrations, screw and rubber-band assembly, and file cleaning
02/04/26	Thursday	Panel File Preparation	Converted fridge magnet files into panels
07/04/26	Tuesday	Drawstring Bag Manual	Completed final manual for drawstring bag
08/04/26	Wednesday	Fridge Magnet Manual	Cleaned illustrator files and developed manual
09/04/26 – 15/04/26	Thursday–Wednesday	Documentation & Assembly	Worked on manuals, BOM, illustrator files, and assembly
16/04/26 – 17/04/26	Thursday–Friday	Appuworld	Worked on Appuworld
18/04/26	Saturday	Mentor Meeting & LDR Project Start	Met mentor, received new project, started LDR research and Appuworld samples
20/04/26 – 02/05/26	Monday–Saturday	LDR Research & Multi-project Research Development	LDR projects, redesigned auto-rickshaw manual, developed fortune wheel concept, updated fridge magnet assembly, brainstormed buzzer-based kit concepts
04/05/26	Monday	Timer Gear Setup	Designed timer setup with gears and explored alternate concepts
04-12/05	Monday - Tuesday	LDR sensor project	Been working on the POC of the Gramophone, Timer, and fortune wheel

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Woxsen University

Woxsen's School of Arts and Design, nestled on a sprawling 200-acre campus in Hyderabad, stands as one of the top design colleges in Telangana. Renowned for its world-class infrastructure, the school provides dynamic spaces that nurture creative thinking and collaboration. Classrooms, labs, and workshops are meticulously designed to global standards, ensuring a comprehensive and enriching learning experience.

Woxsen's faculty boasts a reputation for excellence, comprising renowned professors from top international design institutes and industry professionals. These experts equip students with the essential skills required to excel in the global design industry. The Bachelor of Design (Hons.) program emphasizes the acquisition of knowledge, skills, and creative problem-solving abilities, while the Ph.D. program offers a platform for advanced research and domain expertise.

Woxsen's curriculum seamlessly integrates skill development, design projects, field experience, and theoretical learning, fostering a strong connection between academia and industry. With its cutting-edge infrastructure, passionate mentors, and inspiring learning environment, Woxsen is dedicated to transforming design aspirations into tangible realities. Its unique and progressive approach positions it as the premier design college in Hyderabad, paving the way for future artists and designers.



Indian Industrial Design

B. Des (Hons.), Bachelor of Industrial Design, is an eight-semester program spanning four years. It equips students with a strong foundation in product design, form development, materials, and manufacturing processes. The curriculum introduces learners to human-centered design, ergonomics, systems thinking, sustainability, and aesthetics—essential principles for creating meaningful and functional products.

Students engage in immersive studio sessions, hands-on prototyping, 3D modeling, and design research, with a strong emphasis on human centered design and problem-solving. The program is structured around design projects, industry collaborations, workshops, and internships, providing students with real-world exposure and preparing them for dynamic roles in the design and manufacturing sectors.

Upon completing the four-year program, students graduate with a Bachelor of Design (Hons.) in Industrial Design from Woxsen University, a renowned design school in India. This degree equips them to shape the future with innovative and impactful design solutions.



Synopsis

This journey traces the evolution of a design student from initial curiosity to a more defined and intentional practice. Beginning with limited understanding and a sense of uncertainty, it gradually developed through exploration, reflection, and increasing engagement with the design process.

Over time, experiences across projects, feedback, and continuous learning shaped a more structured approach towards problem-solving and making. What started as experimentation grew into a deeper focus on skill-building, critical thinking, and designing with purpose.

By the final phase, this journey reflects a shift not just in output, but in mindset, towards clarity, responsibility, and confidence in approaching design challenges. It represents a transition from learning what design is, to understanding how to think and act as a designer.

About OiMake

OIMake (pronounced O-I-Make) began with the endless curiosity of a 9-year-old child, who had a lot of questions about how her everyday electronic gadgets worked. Questions like, "How does a radio work?" and "Why do you need Bluetooth on your headphones?" sparked a desire in us to help her understand by doing, not just telling.

As a teacher, I wanted her to explore and discover these answers hands-on. We started with simple electronic circuits, creating light-up birthday cards and figuring out how Bluetooth circuits function. Through this journey, we realized the immense potential in helping other children learn by making and building things from scratch.

This experience inspired OIMake (formerly MakerVerse), a place where kids can dive into the world of electronics and technology. Here, they can grasp both simple and complex science concepts while enjoying the thrill of creating something with their own hands. Our journey started with one child's curiosity, and now we're on a mission to spark that same curiosity in many more young minds.



For busy parents eager to introduce their kids to everyday concepts like Bluetooth, sensors, and motors, we created the perfect solution: a "one-stop shop kit." These kits enable children to build something fun and lasting from scratch while understanding the underlying complex concepts.

Our all-inclusive kits encourage hands-on learning, complete with tools like screwdrivers and spanners, often giving kids their first experience with assembling components. The detailed instruction booklets guide them through the assembly steps and provide troubleshooting tips, fostering an environment of experimentation and learning from mistakes.



Creating their own Bluetooth Speaker, personal Emojibot, or Smart Piggy Bank from scratch instills a profound sense of accomplishment in children. This hands-on process boosts their confidence, enhances problem-solving skills, and builds resilience—all while providing endless hours of fun with their custom-built device!

With this approach, we aspire to cultivate a new generation of makers in India and across the globe!

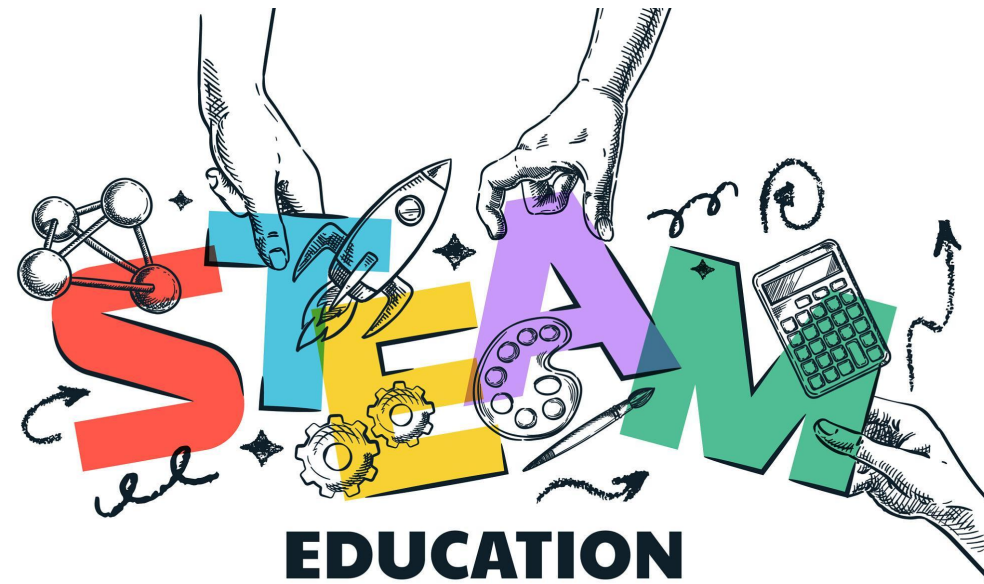
We are proudly ***"Made in India"*** and aim to inspire makers across the world!



Introduction & Context

The STEAM Education

STEAM education combines Science, Technology, Engineering, Art, and Mathematics to encourage creative, practical, and problem-solving skills through interconnected learning. The inclusion of Art highlights the importance of creativity and design thinking alongside technical knowledge. Today, STEAM learning has expanded beyond classrooms into educational toys and DIY kits that allow children to build, paint, experiment, and learn through hands-on interaction. The Indian educational toy market is rapidly growing, with increasing demand for affordable and engaging STEAM products. While many advanced DIY and robotics kits exist, most innovative options are priced above ₹1,500, leaving a gap in the affordable market segment. This project aims to bridge that gap by creating an interactive, technology-integrated DIY experience that combines creativity, learning, and play within an accessible price range for children.



The Screen Time Problem — A Design Opportunity

One of the most consistent and emotionally charged concerns observed during primary research was parental anxiety around children's screen time. Smartphones and tablets have become the default entertainment and activity medium for many children, and parents — particularly urban, educated parents — express genuine concern about the cognitive, social, and developmental implications of excessive passive screen consumption.

This anxiety is a significant behavioral driver in the toy purchase decision. A product that can credibly position itself as a screen-free, engaging, skill-building alternative to phone time immediately addresses one of the most pressing anxieties in its target parent segment. The LED LightBag delivers a 45–60-minute, fully tactile, hands-on building experience with no digital component whatsoever — and produces a tangible result the child can use and display. This is precisely the kind of product a parent wants to see when they walk into a toy store or browse online.



Secondary Research

Landscape of existing products

Each category has developed a distinct user proposition, and each carries specific shortcomings when evaluated against the goal of independent, embedded learning.

Construction kits, magnetic tiles, interlocking gear sets, and model-making systems, emphasis spatial reasoning and fine motor skill. They are generally safe, culturally well-accepted in India, and developmentally appropriate. However, their learning outcomes plateau quickly. Once a child has assembled the kit as directed, there is little incentive for re-engagement unless the child independently shifts into open-ended building, which most do not.

Coding and logic kits have gained significant traction over the last five years, driven by a global emphasis on digital literacy. The limitation here is significant: meaningful engagement typically requires adult facilitation, and the learning curve is steep enough that many kits are abandoned before a satisfying result is achieved.

Robotic kits sit at the intersection of the previous two. Pre-built robots offer the highest visual immediacy but the lowest learning depth, the child interacts with a finished product rather than building and understanding one. Buildable robotic kits offer more, but the assembly is frequently complex, and the feedback mechanism is often limited to a few pre-programmed behaviour.





Indian Market

India presents a distinct context for STEM toy design. Price sensitivity is high: the premium segment is served by international brands at price points that exclude most households outside Tier 1 cities. The affordable domestic segment offers products of variable quality, often without clear pedagogical framing. Educational toy purchases in India are strongly parent-driven, which means the product must communicate learning value clearly at the point of purchase, parents are not buying play; they are buying preparation.

Curriculum alignment matters significantly in the Indian context. Toys that can be linked; even loosely, to school science topics such as circuits, sensors, and energy sources gain immediate traction with parents. Accessibility in terms of assembly language, documentation in regional languages, and the absence of fragile or expensive components are additional factors that determine whether a product succeeds beyond metro retail.

Primary Research

Research Approach

Research was conducted through structured observation sessions with children aged 8 to 12. Observations took place across three environments: a school design club, a domestic home setting, and an after-school activity centre.

The goal was not to test a prototype at this stage, but to understand the natural arc of a child's engagement with unfamiliar objects, their tolerance for instructional friction, and the moments at which curiosity peaks and wanes. Parent perspectives were gathered to understand purchasing motivations and expectations from an educational toy.





Behavioural Analysis

Children in this age group consistently demonstrate a pattern of immediate physical exploration before consulting any instruction. When given an unfamiliar object, the first response is to touch, rotate, or attempt to disassemble it . This tactile-first behaviour is developmentally expected and should be treated as a design input rather than a challenge to manage. A toy that rewards this instinct; that reveals something interesting when handled freely begins building trust with the child before any instruction has been processed.

Children also showed a strong tendency toward comparison and social play. When two children were given the same kit, the primary activity shifted from assembly to negotiation: whose moved faster, whose reacted differently, why was one different from the other. This suggests that variability in the device's behaviour something a calibration-based system naturally produces is a social asset, not an inconsistency to be engineered away.



Target User Psychology — Indian Parents

The true customer of the is not the child but the parent. Secondary research into Indian parental buying behaviour around educational toys reveals a consistent decision framework:

Parent Evaluation Factor	Research Insight
Engagement Duration	Parents seek toys that keep kids occupied for 45-90 minutes — extended, focused play
Screen Time Anxiety	Studies (BMC Psychology, 2025) confirm excessive screen time linked to developmental delays, cognitive issues, and reduced academic performance — parents actively seek alternatives
Learning Proof	Parents want visible, explainable learning outcomes (electronics = circuits; painting = motor skills + creativity)
Age Appropriateness	Clearly defined age labels critically influence purchase confidence
Value for Money	INR 500–1,200 is the sweet spot for enough to trust, affordable enough to try'
Type of Toy	Preference for hands-on, build-it-yourself over passive play toys



Child Development — Benefits of Hands-On Making

The learning outcomes embedded in the DIY STEAM toys are backed by substantial developmental research:

- **Memory Retention:** Students retain 75% of what they learn through practice and 90% when they immediately apply knowledge (National Training Laboratories)
- **Cognitive Development:** Manipulating objects and materials strengthens neural pathways, improving retention and academic performance
- **Fine Motor Skills:** Activities requiring precise movements — painting, inserting components, threading — develop crucial fine motor control
- **Critical Thinking:** Navigating assembly challenges requires problem-solving, assessment, and decision-making
- **Emotional Reward:** Completing a physical build provides a documented sense of achievement and pride, reinforcing goal-setting behaviour
- **Creativity + STEM Integration:** Combining art with science creates simultaneous engagement of both analytical and creative brain functions

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Graduation Project Introduction

The products I am developing fall under the **STEAM DIY Toy** segment — a growing category of hands-on learning kits designed for children that blend Science, Technology, Engineering, Art, and Mathematics into a single making experience.

STEAM toys move beyond passive play. They put real materials, real components, and real processes in a child's hands and ask them to build something from scratch. The learning happens through the making, not through instruction.

The two projects developed under the **OiMake** brand sit firmly within this space.

The **LED LightBag** is a DIY kit where children aged 8 and above paint a drawstring bag, assemble a 6-LED circuit board, and carry a glowing bag they built themselves to school.

The **Light Up Fridge Magnet** is a DIY kit where children stack seven MDF layers, insert a custom PCB with five LEDs, and press a button to light up a mandala they assembled from scratch.

Both products combine art with electronics, deliver a functional object the child keeps, and are completable independently — no adult assistance required.

PROJECT 1



LED Light Bag

Project Overview

The LED LightBag project is a self-contained product design exercise that spans the full spectrum of product development — from identifying a gap in the market to placing a finished, packaged product in a customer's hands. It is categorized under the DIY STEAM toy segment, a rapidly growing class of educational consumer products that use play-based building activities to develop science, technology, engineering, art, and mathematics skills in young learners.

The product takes the form of a drawstring bag — a format chosen for its familiarity, wearability, and ideal canvas-like front surface. The bag arrives with a thematic illustration pre-printed on its front face. The child paints the illustration using the included fabric paints, waits for it to dry, then assembles the five-LED PCB from the kit and slides it into the translucent pocket built into the rear of the front panel. When the battery is connected and the switch is flipped, the LEDs glow through the fabric, illuminating the child's own painted artwork from behind.



Project Synopsis

The LED LightBag is a DIY STEAM (Science, Technology, Engineering, Art, and Mathematics) kit designed for children aged 8 and above. It is built around a simple but powerful concept: give a child a blank canvas in the form of a natural cotton drawstring bag, let them paint it, and then let them make it glow by assembling a five-LED electronic circuit that sits inside a purpose-built pocket behind the painted surface.

The product emerged from a clear observation — the Indian toy market had no affordable, wearable, hybrid kit that combined artistic creation with hands-on electronics assembly for young children. Basic art kits offered painting but no technology. Electronics kits offered circuitry but no creative or personal output. The LED LightBag was designed to occupy that gap: a single, cohesive, 45–60-minute experience that delivers both.

The project was executed in full as part of a graduation internship. It covered every stage of the product lifecycle — from market research and user observation through concept development, form factor selection, electronics design, material testing, illustration development, instruction manual creation, and user testing with real children — culminating in a commercially produced and sold product priced at ₹893, available on the company website and deployed in workshops that have reached approximately 100 children.



Project Objective

The LED LightBag project was initiated with a clear set of objectives that guided every design decision across the development process. These objectives were not simply aspirational — they were treated as measurable benchmarks against which the product was evaluated at each stage of the design thinking process.

Primary Objectives

- | | |
|----|---|
| 01 | Design a product that authentically bridges two learning domains — creative art and basic electronics — within a single, cohesive kit experience, without either domain feeling like an afterthought. |
| 02 | Create a wearable, functional output that a child is genuinely proud of — not just a science experiment to put on a shelf, but something they would carry to school or show to friends. |
| 03 | Target an engagement window of 45–60 minutes of focused, hands-on, screen-free building activity that holds a child's attention through variety: painting, drying, assembling, and the dramatic reveal. |
| 04 | Design for independence — a child aged 8 and above should be able to complete the entire kit using only the included visual instruction manual, with no requirement for parental assistance. |
| 05 | Develop an electronics assembly that is child-safe, low-voltage, and intuitive enough that a child can complete it correctly without risk of error or injury. |
| 06 | Fill a clear market gap — no comparable product in the accessible Indian toy market (below ₹1,000) combines art-making and electronics assembly in a wearable form. |

Project Brief

DIY backpack kit that combines creative expression and basic electronics, allowing kids to design and buildt their own illuminated bag. "

Competitive Landscape — Existing DIY & Light-Up Kits

An analysis of global and Indian market players reveals a clear product gap. No existing kit on the Indian market combines wearable textile art (fabric painting) with basic electronics assembly in a single child-accessible product.

Market Gap Summary

The research confirms a strong, unfilled product gap: no affordable Indian product combines fabric art with a guided LED electronics assembly for children aged 8+. Existing products either offer art without tech, or tech without art, and those that combine both are either adult-targeted, require coding/soldering, or are priced far above the target segment.

Product / Brand	Price Range	Art + Electronics?	Gap Identified
Smartivity STEAM Kits (India)	INR 500–1,500	No — mechanical only	No textile/art component
Adafruit Flora / GEMMA (Global)	USD 20–50	Yes, advanced	Requires coding; not child-friendly
SparkFun LilyPad Kit (Global)	USD 15–40	Yes, advanced	Soldering/coding required
Lumen Couture DIY LED Kit (Global)	USD 30–60	Partial	No painting; adult-focused
TechnoChic Twinkling Tutu (Global)	USD 25–45	Yes	No painting; sewing-heavy
360Fashion LED Bag Kit (Global)	USD 500+	Yes	Premium; not for kids
Generic Coloring Bag Kits (India)	INR 200–600	No — coloring only	No electronics whatsoever
LED LightBag (Your Product)	INR 893	YES — unique combination	Gap FILLED

Market Products images

sl. No	link	product and price	recent sales	images
1	https://amazon.in/Bloomingo-DIY-Kitty-Bag-Kit-for-Girls/	₹444 Bloomingo DIY Kitty Bag Kit for Girls/	50+ last month	
2	https://www.amazon.in/Arishto	₹999 Arishto DIY Bunny Crochet Bag Kit for Kids	no info	
3	https://www.amazon.in/GIGGL	₹299 Sewing Craft Kit for Kids Handmade Educational Project Random Colour Pack 1	no info	
4	https://www.amazon.in/-/hi/PO	₹199 PokuPop DIY Fancy Kids Sewing Bag Kit – Sew Your Own Felt Tote, Creative Craft Activity for Boys & Girls	no info	
5	https://www.amazon.in/HASTHIP	₹561 HASTHIP® DIY Handmade Bag Material Kit Girls Purse Bag Hardware Chain Design Shoulder Bag Material Kit DIY Lace Weaving Bag Material Kit with Tools & Accessories DIY Bag Gift for Girls	4.2* (53 global	
		₹668 HASTHIP® Plastic Mesh		

Market Products images

sl. No.	form of the bag	platform and link	product detail	material	images
1	sling bag	deo dap -- https://deodap.in/products/mermaid-tail-shaped-kids-sling-bag-1-pc?variant=53042824806710&country=IN&currency=INR&utm_medium=product_sync&utm_source=google&utm_content=sag_organic&utm_campaign=sag_organic&srsitid=AfmBOocMFnVG-KBRetIJFdOw-DBMNn7eb3Sfw8wngjZqmxkZIVd1MyIQE0	29rs	Mermaid Tail St Fabric with Sequins & Synthetic Tail Design	 
2	cross body bag	ARHAMSMART - The Wholesale Mart https://arhamsmart.com/product/kids-crossbody-sling-chest-bag/?attribute_pa_kidschestbag=rainbow-unicorn&srsitid=AfmBOoq_gtDyYiD3iWdUWIZSjuSk8j8ro9QFgjqX0Cqj6F9KFBDxiGVMOfg	₹110	Kids Crossbody Sling Chest Bag	
3	totebag	India mart - https://www.indiamart.com/proddetail/canvas-tote-bag-22889698355.html?utm_medium=prd_ads&utm_campaign=23411315669&utm_ad_group=194444872761&utm_content=product&utm_source=google&gclid=Cj0KCQiApfjKBhCOARIsAMiR_IsB-wdAAmccS1Qc2A7Mk3RAI_iGQCyuBhF9_5XhPXR4qsq8sqQc1eEaAs0QEALw_wcB		Plain Cotton Canvas Tote Bag, upto 7 kg	

Form Factor Ergonomics.

Choosing the bag itself — and sizing it for the body of an eight-year-old child.

Different bag formats such as backpacks, tote bags, sling bags, crossbody bags, and drawstring bags were explored during the concept development stage. After evaluating their usability, aesthetics, and interaction potential, the drawstring bag was selected as the final direction. Its compact form makes it suitable for children, while its **timeless appeal** keeps it visually relevant across generations. Additionally, the **large front surface** provides better opportunities for expressive artwork integration and interactive lighting features, making the product more engaging and playful for kids.

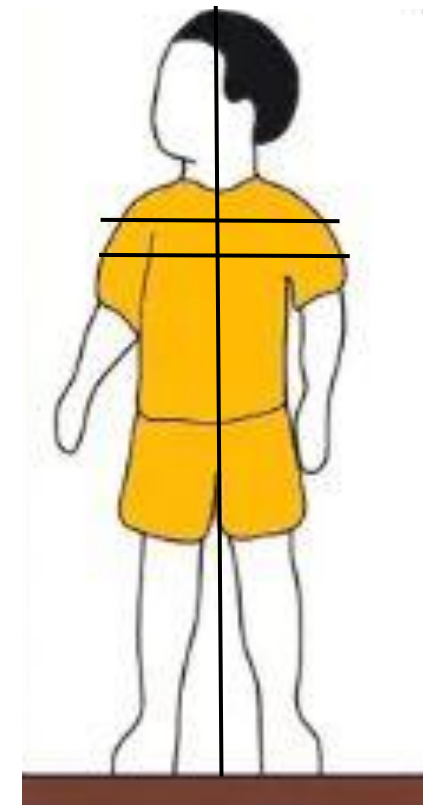


Anthropometric Research — Child Body Measurements

Because the LED LightBag is a wearable product, ensuring it fits comfortably and looks proportionate on a child's body was a design requirement. Standard anthropometric data for children aged 8–14 was studied to inform the bag's final dimensions:

These measurements informed the final bag dimensions, settled at approximately 30cm wide by 40cm tall — a universal fit across the 8–14 age range that provides adequate front panel canvas area without being too large for younger children to carry comfortably.

Shoulder Width — 8 to 10 years	Approximately 28–33 cm
Shoulder Width — 11 to 14 years	Approximately 33–39 cm
Back Length — 8 to 10 years	Approximately 30–38 cm
Back Length — 11 to 14 years	Approximately 38–46 cm
Front Panel Access	Children in this age range have sufficient reach and dexterity to paint the front panel while the bag is laid flat on a table
Component Handling	Children aged 8+ have sufficient fine motor development to handle 5mm LEDs and push them into PCB sockets
Drawstring Grip	Adequate hand strength to pull and tie drawstring closure without difficulty



Material and Painting exploration

The material and painting exploration focused on finding the most suitable surface and interaction method for the children's DIY drawstring bag. Different painting techniques were tested through trial-and-error experiments to understand paint behavior, including absorption, spread control, and drying time on fabric surfaces. The process also evaluated how easily children could paint independently and how clean or messy the activity became during use.

Various materials such as muslin fabric, bleached muslin, non-woven fabric, felt cloth, and cotton surfaces were explored and compared. Each material responded differently to paints, textures, and drawing styles, affecting the overall visual output and user experience.

The exploration aimed to identify a material that was easy to paint on, visually appealing after coloring, safe for children, and durable enough for repeated use. Comfort, flexibility, and paint visibility were also considered as important factors during the selection process.

Different painting mediums were evaluated to identify the most suitable option for the LED LightBag experience. Acrylic fabric paints provided strong and durable colour output but offered lower precision and required better control and drying time, making them more suitable for older children. Fabric markers such as muslin are the most effective medium due to their controlled application, minimal mess, ease of use, and child-safe non-toxic properties. Washable markers were safer but lacked durability after washing, while solvent-based and alcohol markers were avoided because of higher VOC emissions, safety concerns, and increased cost.

Non Woven



Propylene Canvas

Wool Felt



Synthetic Bag

Muslin



Bleached Muslin



Acrylic Paints



Acrylic markers



Fabric Markers



Sketch Pens



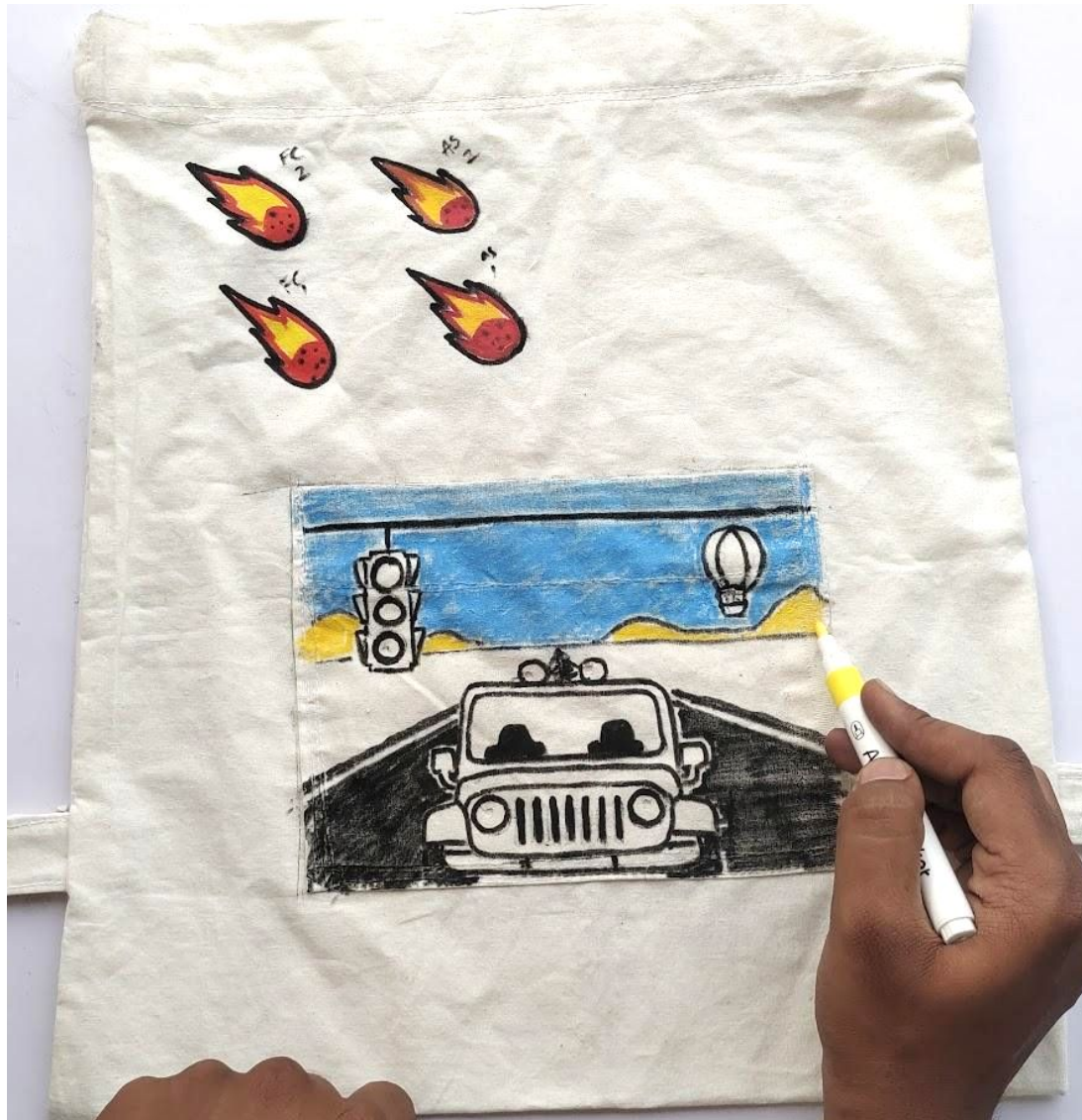
Washable Pens



Paints







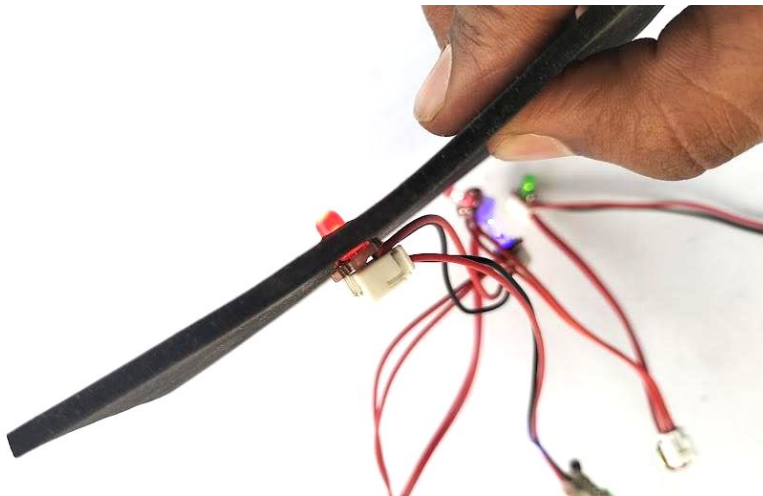
Tech integration

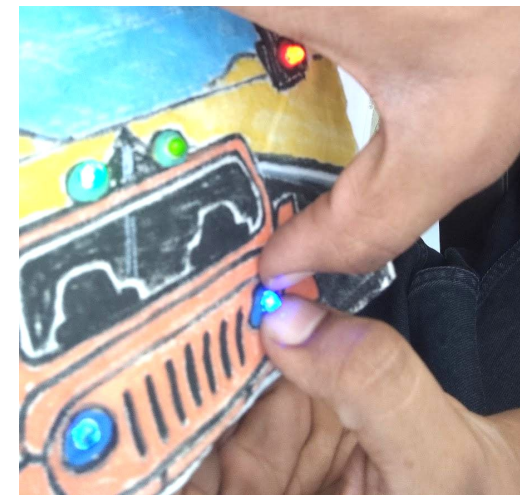
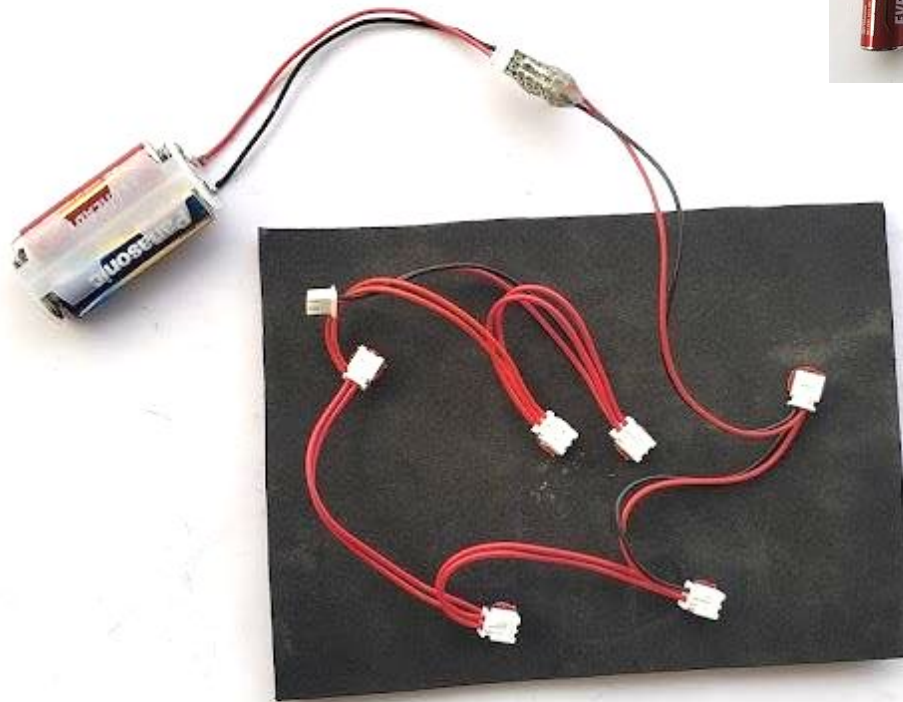
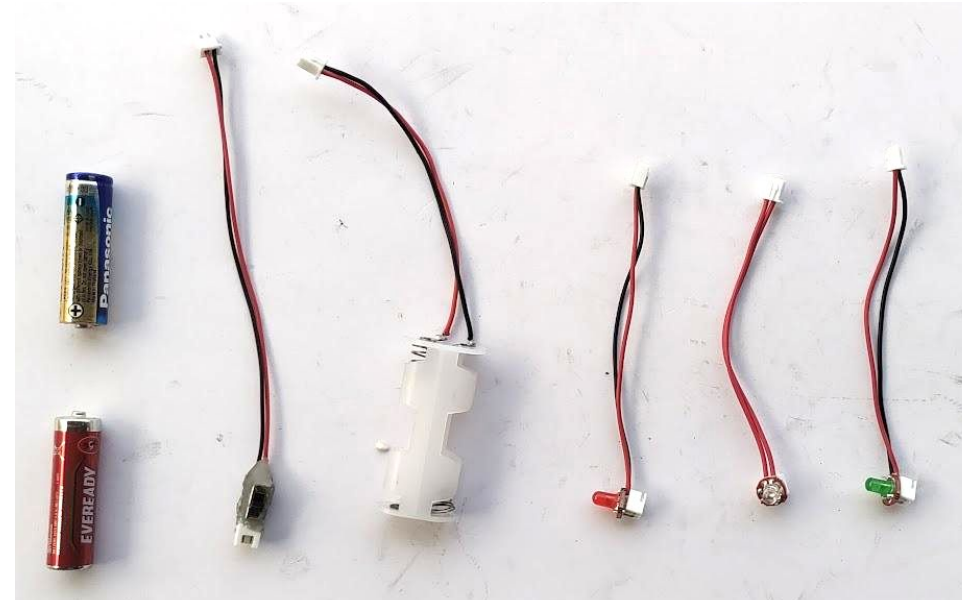
The technology integration research focused on exploring suitable lighting systems and LED integration methods for the LED LightBag. Different types of LEDs, including DIP (Dual In-Line Package), SMD (Surface Mounted Device), and COB (Chip on Board), were studied and compared based on brightness, size, flexibility, and ease of integration with fabric surfaces. Multiple attachment methods such as direct LED attachment, pocket-based systems, and button or PCB spacing mechanisms were also explored to identify the most practical and child-friendly solution.

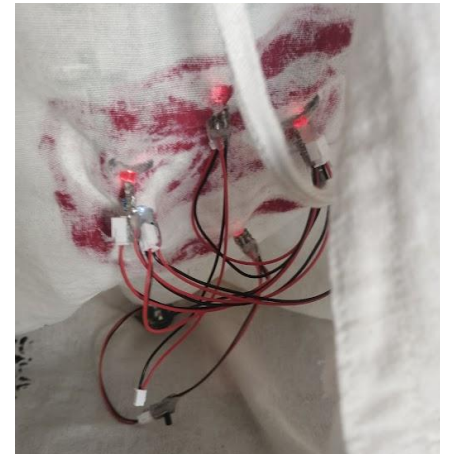


Tech integration

Different methods of integrating the LED assembly into the bag were explored, including direct threading through fabric and adhesive mounting systems. While these methods provided visual appeal, they lacked durability and were difficult for children to assemble safely. The final selected approach was a PCB + EVA + pocket system, where a custom LED PCB slides into a semi-transparent pocket behind the front panel, allowing led to pass evenly through an EVA foam layer and the fabric surface in a safe, durable, and child-friendly manner.







Visual Language & Illustration Research

The visual language and illustration research primarily focused on understanding the graphic styles and visual trends commonly used in children's products through market observation and analysis. Research was conducted by studying story books, cartoon merchandise, DIY kits, and educational toys to identify recurring themes, color palettes, and illustration styles that attract children. The study revealed that children are more engaged by simple, bold, and expressive visuals featuring themes such as animals, fantasy elements, and cute characters. Bright and vibrant colors were found to be more appealing, while dull or muted tones showed less engagement. The findings helped establish a clear direction for creating child-friendly and visually engaging graphics.





Graphic & Layout Development

The graphic and layout development phase focused on experimenting with different illustration arrangements and integrating them effectively with the LED lighting system. Multiple layout formats, including centered and spread-out compositions, were explored to study how graphics interacted with the LED placements beneath the fabric surface. The development process aimed to ensure that the illuminated sections enhanced the overall visual appearance without causing confusion during the painting activity. Special attention was given to maintaining clarity, usability, and visual balance so that children could easily understand and engage with the activity. The overall objective was to create a cohesive experience where the illustration layouts and lighting system functioned together seamlessly.



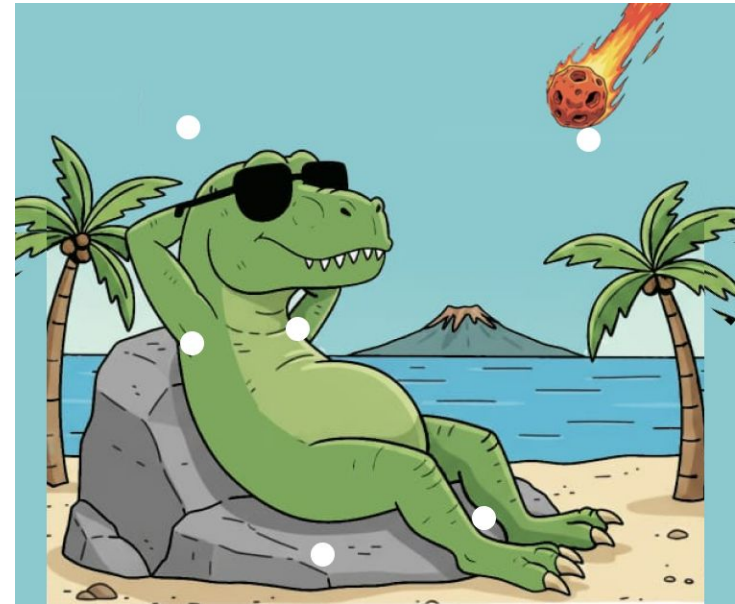
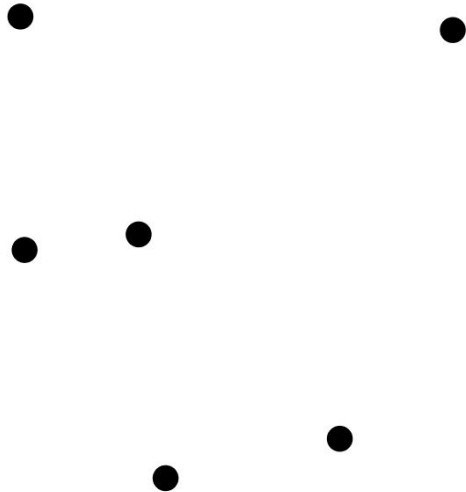


Illustration Development & Visual Refinement

The graphic development process focused on creating engaging and child-friendly illustration concepts tailored for the LED LightBag experience. Multiple illustration ideas were developed and refined through continuous iteration, with emphasis placed on shape simplicity, line clarity, and overall engagement level. To validate the visual direction, observational research was conducted in toy stores and children's product sections, where popular illustration styles, patterns, themes, and compositions were studied and analyzed.

The research revealed that effective children's graphics are bold, expressive, easy to trace, and visually playful without being overly detailed or confusing. Based on these insights, the illustrations were repeatedly refined through prototype testing, evaluating their compatibility with the lighting system and ease of painting for children. The final understanding derived from the process was that the graphics function not only as visual elements but also as interactive guides that support creativity, painting activity, and light-based engagement simultaneously.

Vehicle



Places



Animals



Objects



Emotions



Comic text



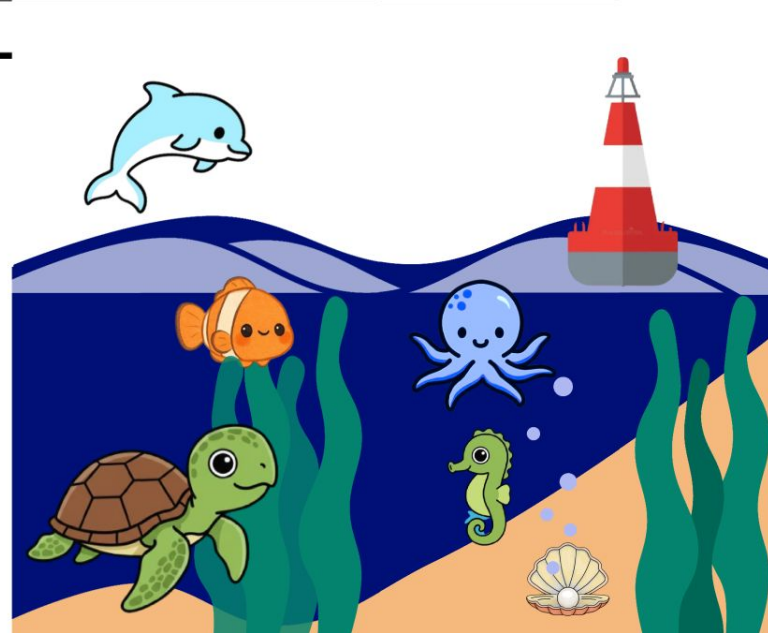
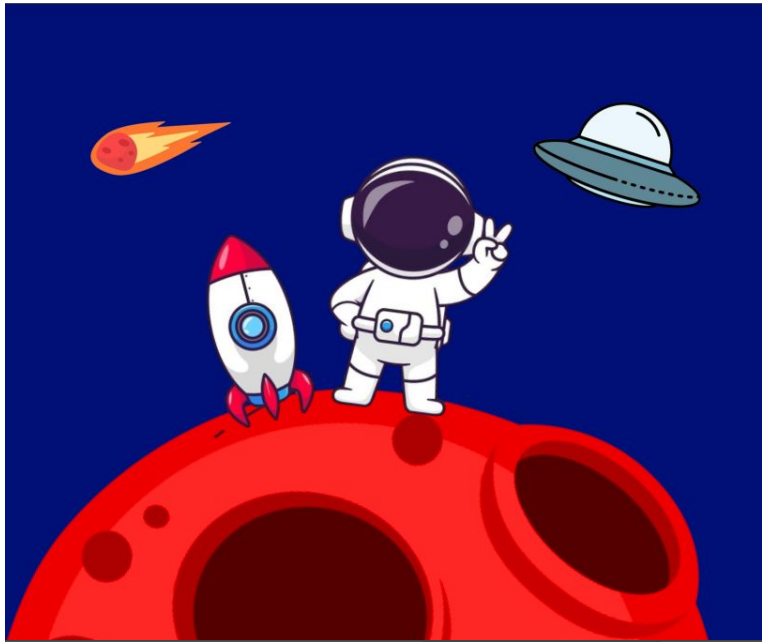
Tasks

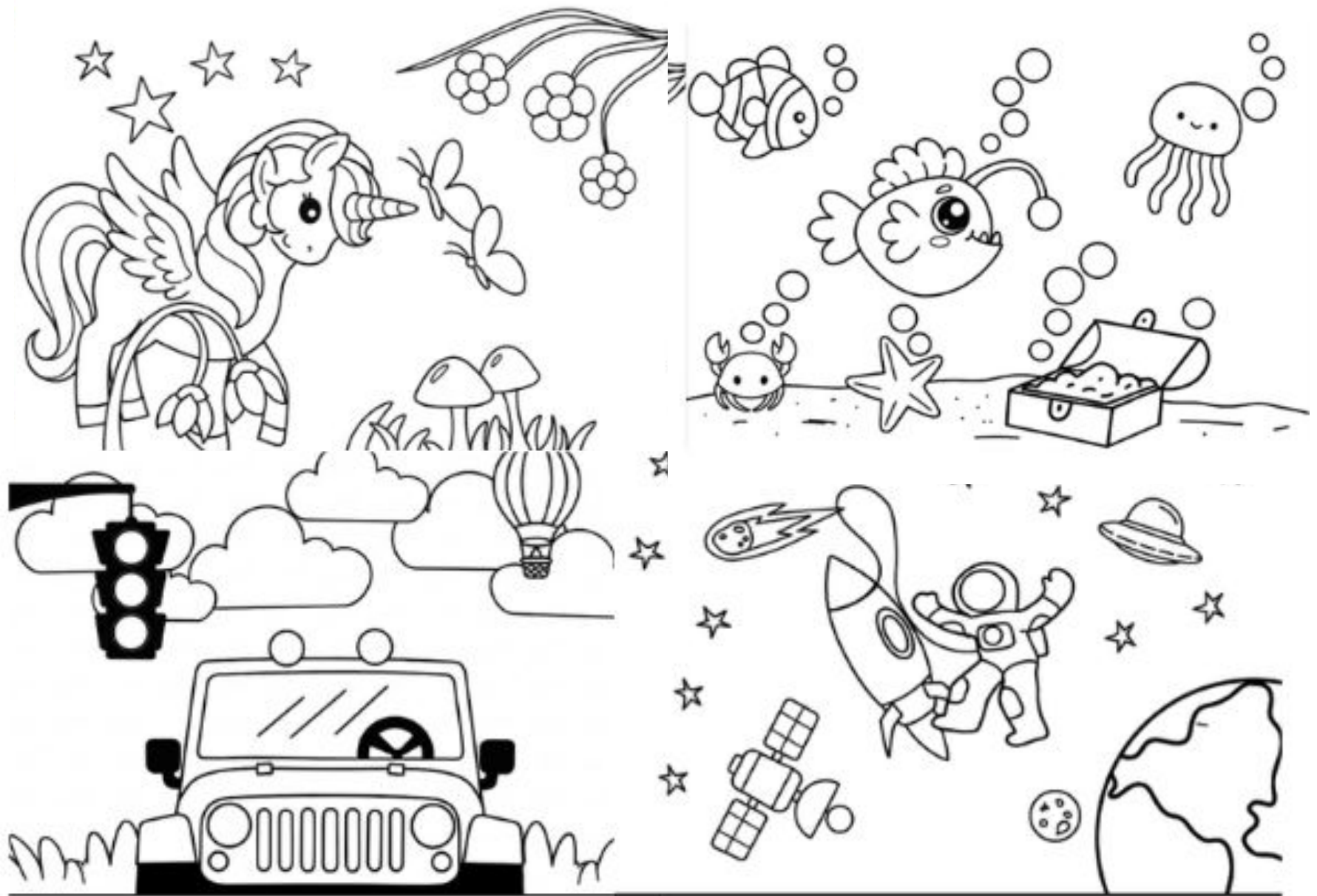


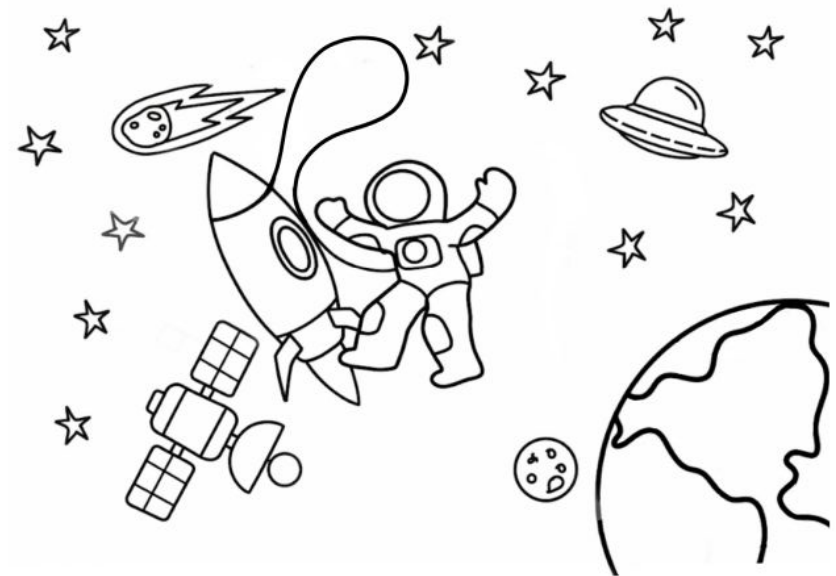
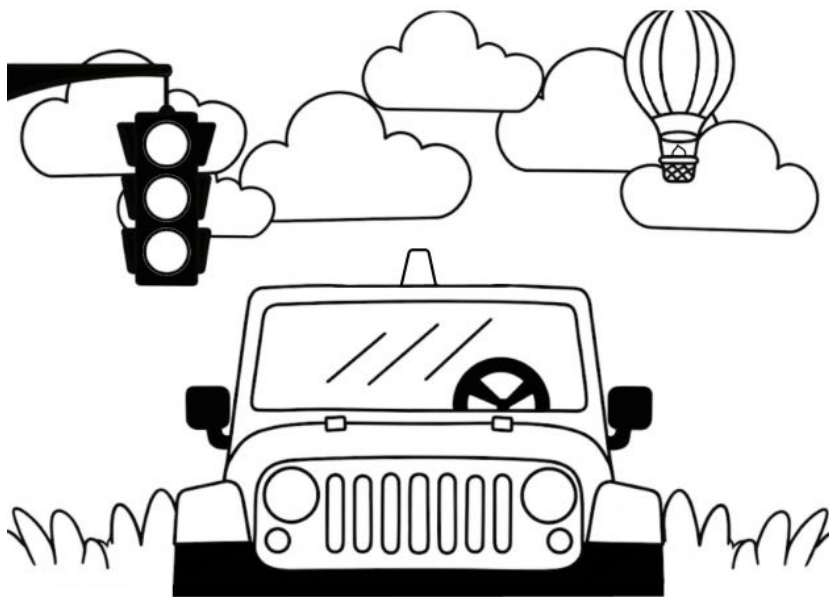
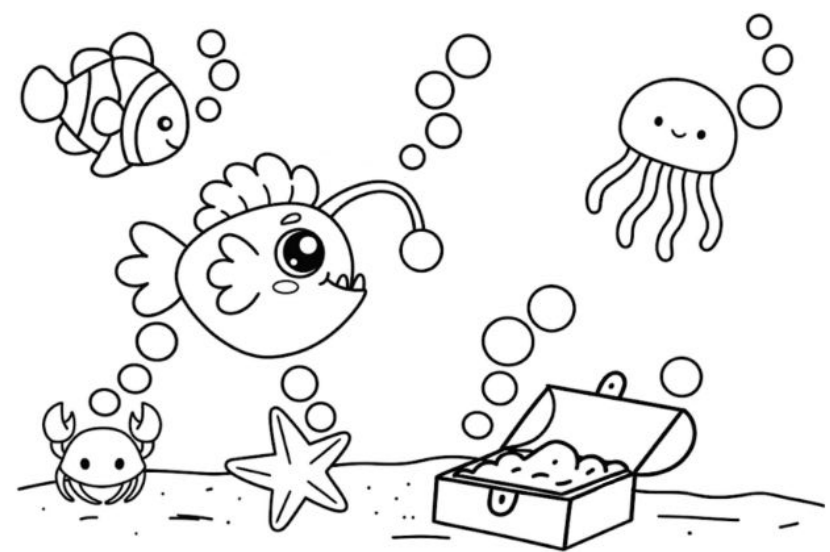
Missilinous











Functional Prototyping & Lighting Evaluation

Physical prototypes of the LED LightBag were developed to test the integration of the fabric structure, pocket-based lighting system, and LED setup within a functional product experience. The prototyping phase focused on understanding how effectively the lighting system interacted with the illustrations and how easily the components could be assembled and used by children. Different configurations of LED placement and pocket arrangements were tested to achieve a balanced and visually engaging glow effect across the artwork.

Key evaluations included:

- Light distribution across the illustration surface
- Battery performance and runtime
- Visual enhancement created by the glow effect
- Whether the lighting appeared too harsh or too dim
- Ease of installing LEDs into the pocket system
- Component fit and overall assembly feasibility

The testing revealed that light placement needed to align carefully with the illustration elements and remain evenly distributed for a stronger visual impact. Additionally, the system had to remain simple, safe, and intuitive enough for children to assemble and interact with comfortably.



User testing & Experience Testing

Hands-on workshops were organized with children to evaluate the usability, engagement, and overall interaction experience of the DIY LED backpack kit. During these sessions, children were provided with the complete kit along with a photo-based instruction manual designed to guide them through the activity in a simple and understandable manner. The workshops helped observe how children interacted with the product independently and how effectively they understood the assembly and customization process.

Children were encouraged to:

- Paint and personalize the backpack illustrations
- Assemble the lighting system independently
- Follow the photo manual step-by-step
- Complete the overall DIY experience with minimal guidance

The workshops provided valuable insights into user behavior, ease of assembly, creativity levels, and the effectiveness of the instruction system. The sessions also helped identify areas where the process could be simplified further to create a more intuitive and enjoyable experience for children.







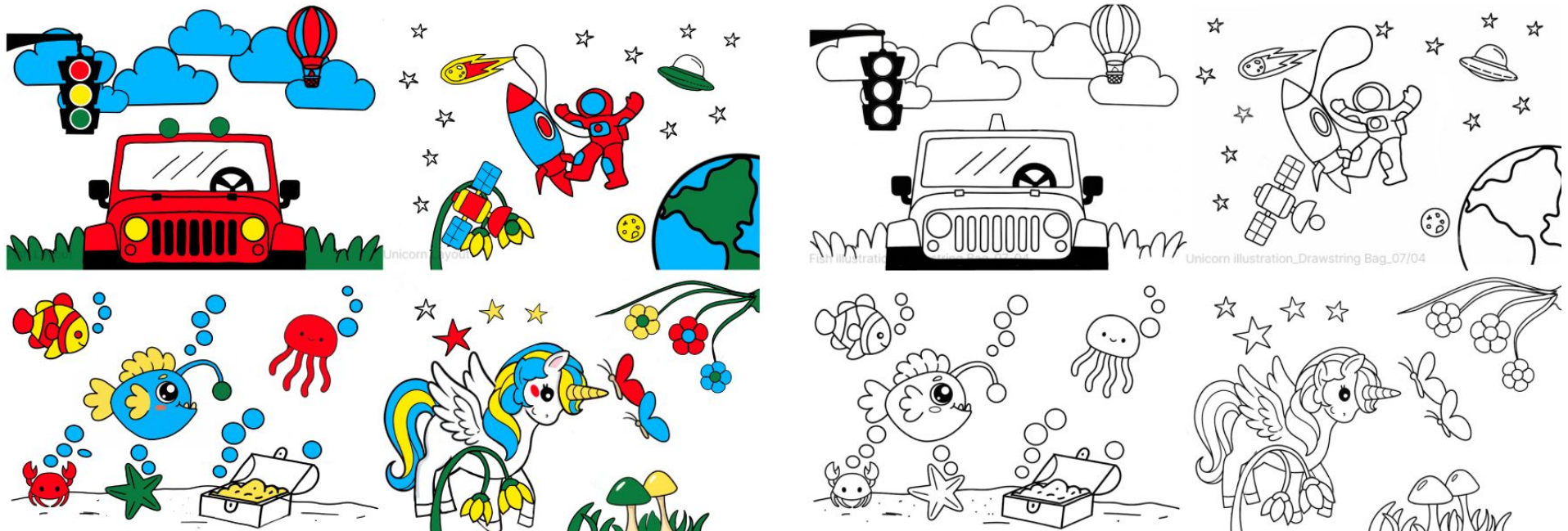
Feedback Integration & Design Refinement

Insights gathered from prototype testing and children's workshops were used to refine and improve the overall DIY LED backpack experience. The feedback helped identify areas where children faced confusion, difficulty, or reduced engagement during painting and assembly. Based on these observations, multiple aspects of the product were simplified and optimized to create a more intuitive, enjoyable, and child-friendly interaction flow.

Key refinements included:

- Simplifying illustrations for easier tracing and coloring
- Optimizing color usage to reduce visual clutter
- Reducing lighting system complexity for easier assembly
- Improving photo-based instructions for better understanding
- Enhancing the overall user flow and activity sequence

These refinements helped improve usability, reduce assembly errors, and create a smoother and more engaging creative experience for children.



Instruction Manual Development

A step-by-step instruction manual was developed for the DIY kit with a strong focus on visual communication rather than text-heavy explanations. The manual was designed to be sequential, intuitive, and easy for children to understand independently. Real process photographs were converted into simplified illustrations to reduce visual noise, improve clarity, and create a more universal and child-friendly instruction system.

The manual included:

- Overview of kit components
- Step-by-step assembly guidance
- Coloring and customization instructions
- Basic safety instructions

This approach helped make the overall DIY experience more engaging, accessible, and easier for children to follow with minimal supervision.



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~Explore ~Experiment ~Experience ~Electronics



LED Light Bag

DIY Guide

Version 1.1



 Oimake

1

LED Light Bag
Oimake

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+91 70750 15103

Explore - Experiment - Experience - Electronics

Warnings and Caution
Read the below instructions carefully before using the product.

Warning! Contains small parts. Not Suitable for Children Under 3 Years of Age.

The product uses 2 x AA cells of maximum 1.5V each. Do not exceed a total voltage of 3V DC.

Recommended Battery types: Dry cell, Alkaline cells, NiMH.

- Non-rechargeable batteries are not to be recharged.
- Rechargeable batteries are only to be charged under adult supervision.
- Rechargeable batteries are to be removed from the toy before being charged.
- Different types of batteries or new and used batteries are not to be mixed.
- Batteries are to be inserted with the correct polarity.
- Discharged batteries are to be removed from the toy.
- The supply terminals are not to be short-circuited.

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USE COUPON CODE - **MAKER25**

What's in the box

Acrylic Markers
Markers - 4
Pack 1

Carbon paper
Pack 2

LED Lights - 5
Paper Clips
Battery Holder

STEP 1

*Make sure the yellow side of the carbon paper is facing up (towards the paper).

STEP 2

Use a blunt tip pencil to trace.

*Pencil not included in the box

STEP 3

*Remove the paper clips, illustration sheet and carbon paper

STEP 4

*Use Black marker pen to highlight the borders

STEP 5

*Start painting the bag with the colours.

Series Connection

Imagine, all cars moving on one single road. Every car has to follow the same path one after another. If one car stops, it blocks the road and all the cars behind it get stuck.

Similarly, LEDs in series are connected in a line, so current has only one path. If one LED fails, all the LEDs turn OFF.

Also, Voltage gets divided between each LED - Adding more LEDs results in dimming or may not glow.

To solve this problem, We connect Multiple LEDs in Parallel Connection.

Parallel Connection

In a parallel connection, each car have its own path, so traffic moves smoothly. Even if one road is blocked, cars can still move through other roads.

Similarly, LEDs in parallel have separate paths, so each LED works independently. If one LED fails, the others still glow.

Also, Each LED gets full voltage - LEDs glow bright and equally.

This is the reason why you can see all LEDs are glowing at same Brightness.

STEP 6

*requires 2 AA batteries, not included in the box

STEP 7

Check if LEDs are glowing or not.

STEP 8

STEP 9

STEP 10

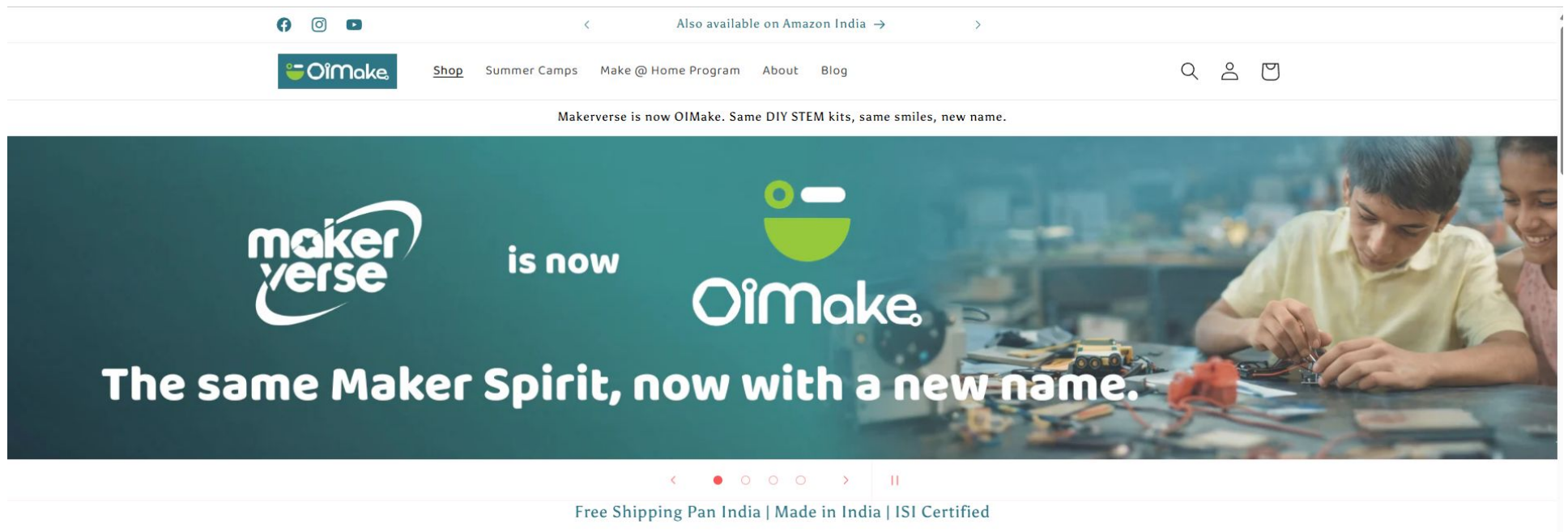
Turn the Switch ON and your Swag tool

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Sleeve Design

Standard box (pre-ordered); branded sleeve printed externally; sleeving and final kit assembly done in-house







Best Seller

Bluetooth Speaker | DIY STEM Kit | 8-14 yrs | Fun with Audio Engineering

8-14 yrs Fun with Audio Engineering

★★★★★ 11 reviews

Rs. 1,699.00 Rs. 1,357.00



Best Seller

Autorickshaw | DIY STEM Kit | 8-14 yrs | Fun with Motors & LEDs

8-14 yrs Fun with Motors & LEDs

★★★★★ 11 reviews

Rs. 1,299.00 Rs. 1,189.00



Best Seller

Projecto | DIY STEM Kit | 8-14 yrs | Fun with Electronics

8-14 yrs Fun with Electronics

★★★★★ 8 reviews

Rs. 1,299.00 Rs. 1,135.00



Sale

Cherry Blossom Lamp | DIY STEM Kit | 8-14 yrs | Art meets Tech

8-14 yrs Art meets Tech

★★★★★ 6 reviews

Rs. 1,299.00 Rs. 970.00



Sale

GloLamp | DIY STEM Kit | 8-14 yrs | Fun with 3-Color Light Circuits

8-14 yrs Fun with 3-Color Light Circuits

wood-finish ★★★★★ 9 reviews



Sold out

DigiSafe | DIY STEM Kit | 8-14 yrs | Fun with Digital Security

8-14 yrs Fun with Digital Security

★★★★★ 8 reviews

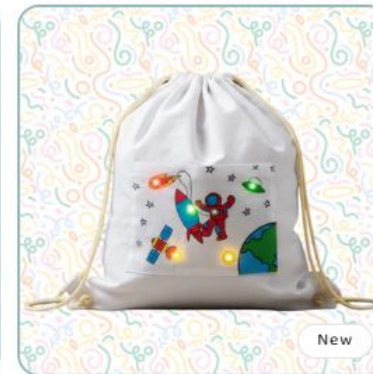


New

TaskChamp | DIY Chore Chart Kit | 8-14 yrs | Build. Track. Glow.

8-14 yrs Build. Track. Glow.

Rs. 1,299.00 Rs. 971.00

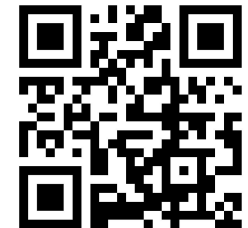


New

LED Light Bag | DIY STEM Kit | 8-14 yrs | DIY Wearable Tech Kit

8-14 yrs DIY Wearable Tech Kit

Rs. 1,099.00 Rs. 893.00



Makerverse is now OIMake. Same DIY STEM kits, same smiles, new name.



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Design it, Light it up & Watch it Glow!

The **DIY LED Light Bag** by OIMake is your child's first wearable tech project. Kids wire up multiple LED circuit boards in parallel, connect a battery, and watch their bag light up — then sling it on their back and wear their creation to school, camp, or wherever the adventure takes them.



Workshops with the kids

The product is ready for the sale in the website and not only purchase but also the bag is used and build by kids in the custom workshops by the company

Almost there are 100 kids right now using the bag









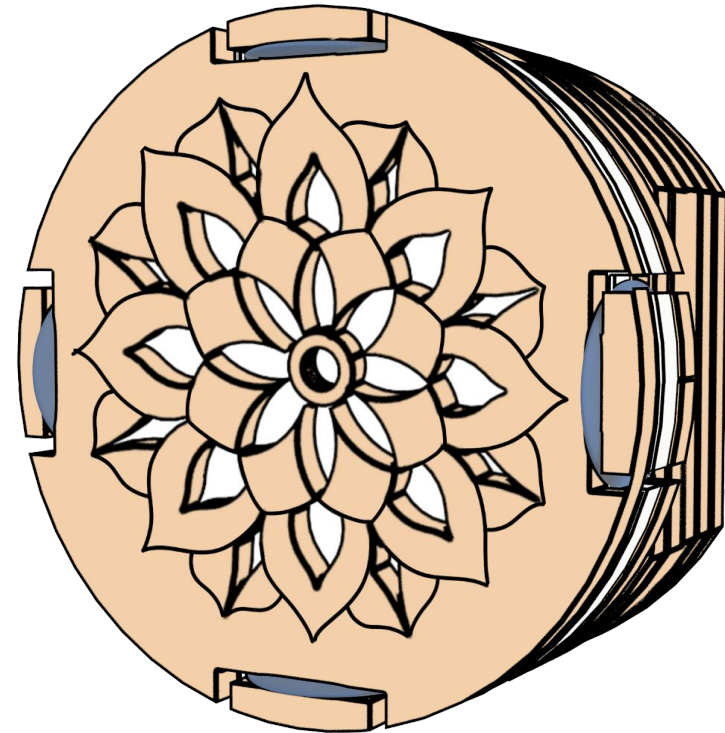
PROJECT 2



Light-up Fridge Magnet

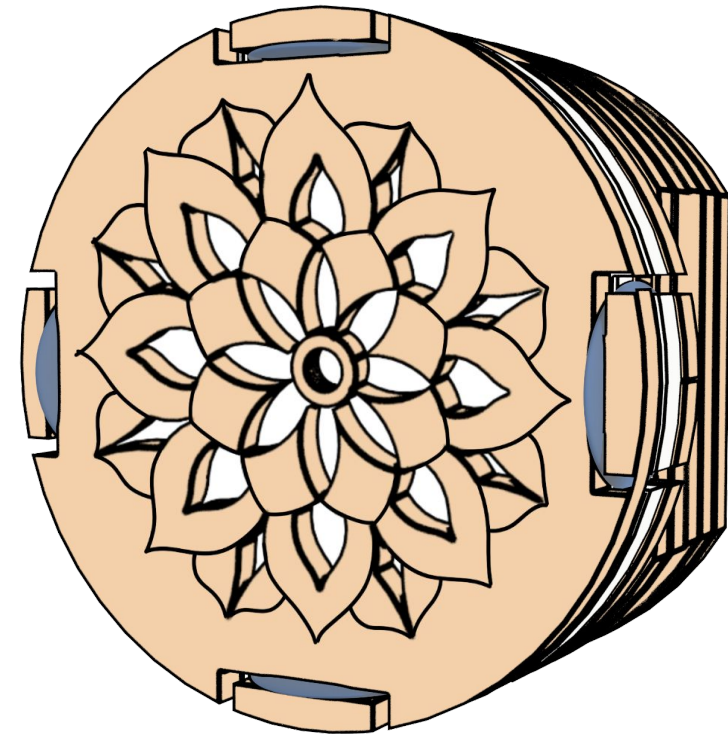
Project Overview

The Light Up Fridge Magnet is a technology-integrated DIY toy kit developed by OiMake for children aged eight years and above. Unlike conventional fridge magnets — which function purely as passive decorative objects — this product introduces an active lighting interaction: the child assembles the kit layer by layer, inserts the electronics, and the completed magnet illuminates through a push-button mechanism powered by a coin-cell battery.



Project Synopsys

The Light Up Fridge Magnet is a do-it-yourself STEM toy kit designed for children aged eight years and above. The product reimagines the passive fridge magnet as an active, child-assembled electronic object — a layered MDF construction housing a custom printed circuit board with five LEDs, a push-button switch, and a CR2032 coin-cell battery, all assembled by the child through ten sequential steps using only a visual instruction manual. The development process progressed through multiple iterative stages across four phases: concept and ideation, electronics development, PCB refinement, and assembly optimisation. Key transitions included upgrading from a single LED to a five-LED array, replacing a slide switch with a push-to-toggle mechanism, iterating PCB dimensions to fit the MDF cavity, and shifting from rubber-band-only clamping to a screw-guided stacking system that reduced child assembly errors significantly.



Project Brief

A child-assembleable DIY kit producing a functional illuminated fridge magnet — in which the child installs the electronics, stacks the MDF layers, and operates a push-button glow effect as the primary user interaction.

Form Exploration

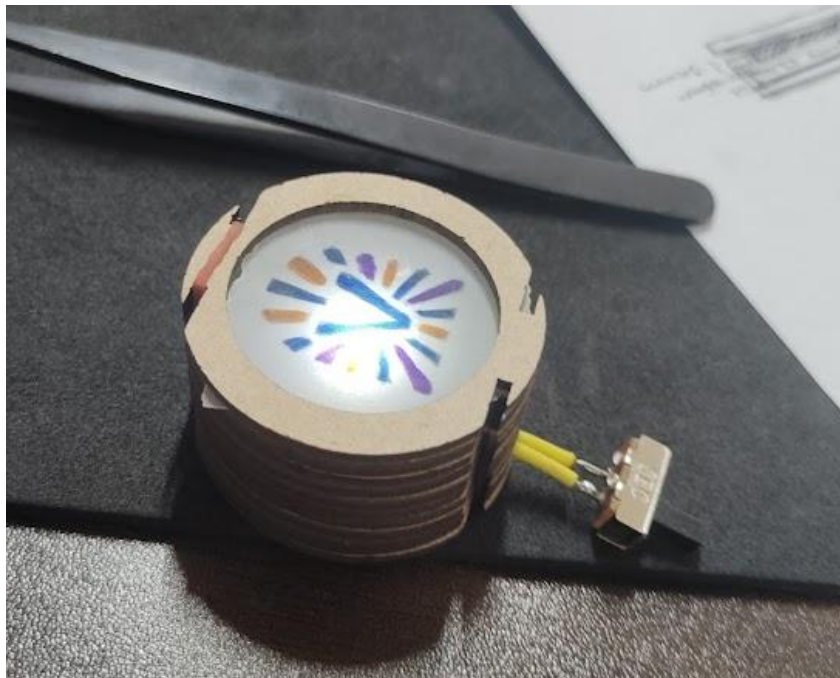
Different visual forms and shape styles were explored during the concept development phase, including geometric, organic, and emblem-based approaches. Each form was evaluated based on visual appeal, recognizability, interaction with lighting, and suitability for children's products. After multiple iterations, a compact circular emblem-style form was finalized, creating a badge-like identity that felt playful, visually balanced, and easy to integrate with the illustration and LED system. The circular form also helped improve focus on the artwork while maintaining a clean and child-friendly appearance.



From idea to first prototype.

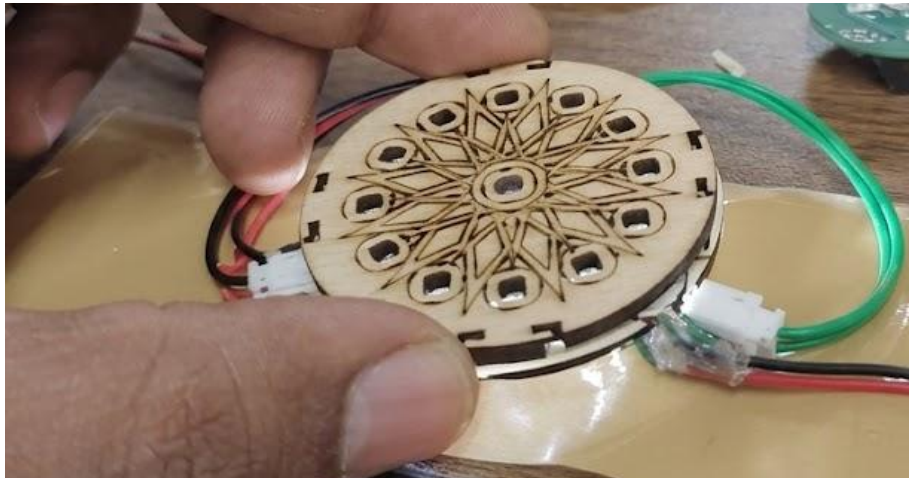
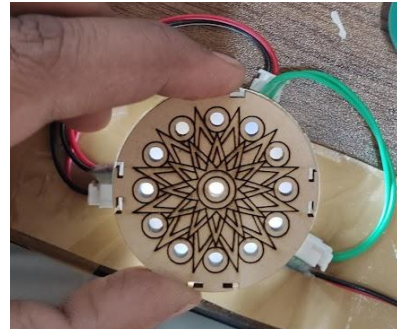
Stage 01 — Idea Generation The concept originated from observing the gap between passive home decorative accessories and the STEM toy gifting market. The hypothesis: a fridge magnet could be reimaged as a child-built, electronics-integrated experience without losing its primary function as a refrigerator decoration.

Stage 02 — Initial Prototype Development The first prototype was a hand-assembled stack of rough-cut MDF sheets held together with tape and rubber bands, with a single LED connected by loose wires to a battery and slide switch. Built to answer one question: can a glowing effect be achieved through an MDF cutout layer?



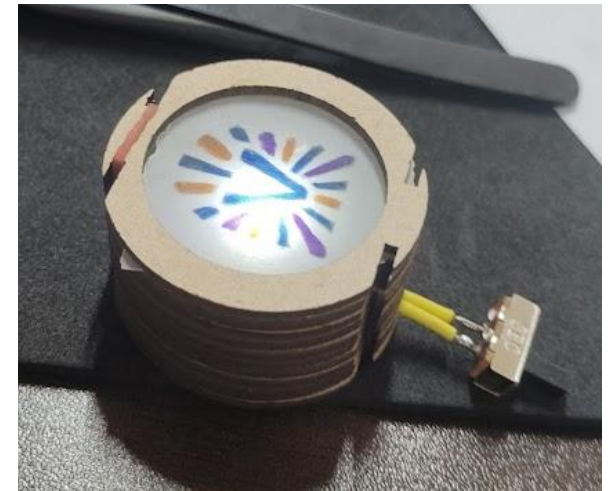
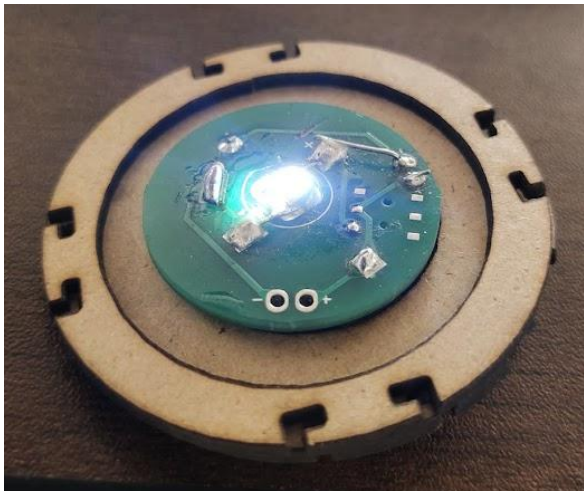
From idea to first prototype.

Stage 03 — Pattern Development With the basic glow hypothesis confirmed, focus shifted to the decorative top face. Multiple mandala-style patterns were evaluated against visual complexity when unlit and quality of glow diffusion through the negative-space cutouts when lit. The final mandala pattern was selected for its balanced density and visual familiarity in home decor contexts



From idea to first prototype.

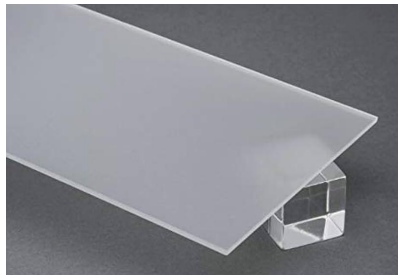
Stage 04 — Single LED Setup Initial configuration used one centred LED. The glow was uneven — petals nearest the centre were substantially brighter than those at the perimeter, producing an unacceptable gradient



From idea to first prototype.

Stage 05 — Diffuser Material Testing Multiple translucent diffuser sheets were tested between the LED and the decorative face: tracing paper, frosted acrylic made different frosted acrylic with different hatching techniques over the laser cutter, and also made a sanded surface over the sheet. Nothing gave us the right dispersion.

Stage 06 — Adjustable Jig for Light Height Testing A testing jig was fabricated to vary the vertical distance between LED surface and the diffuser. Results showed a 6–8 mm gap produced optimal diffusion. This directly informed the MDF spacer depth



From idea to first prototype.

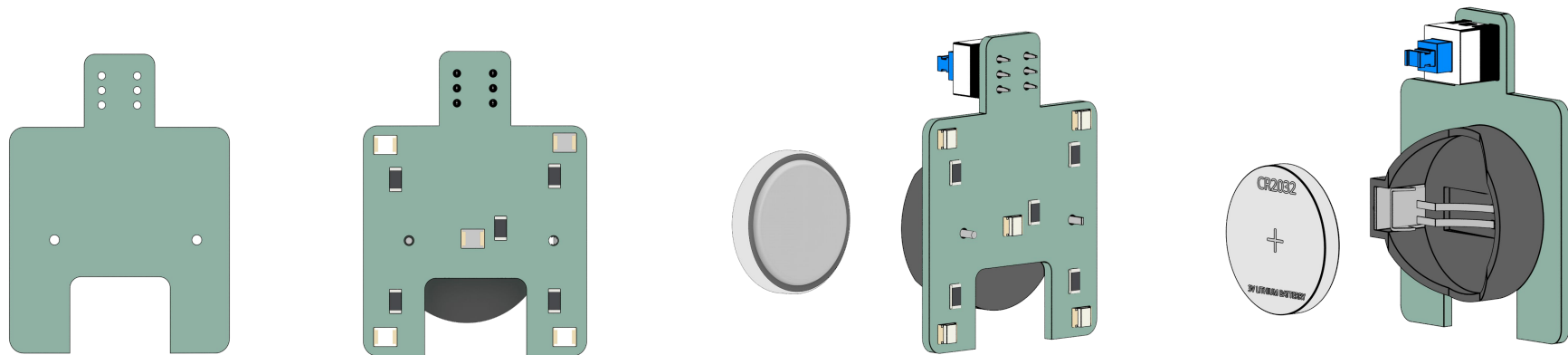
Stage 07 — Upgrading from 1 LED to 5 LEDs The single LED was replaced with a five-LED array at the centre and four angular positions. The combination of spatial distribution and diffuser produced substantially more uniform glow.



Stage 08 — Slide Switch to Push Switch User observation revealed the slide switch direction was non-intuitive and prone to accidental state changes when the product was moved. Replaced with a push-to-toggle button providing clear tactile feedback and resistance to accidental actuation



Stage 09 — Custom PCB Development and Testing A custom PCB was designed to consolidate all electronic components onto one board, eliminating loose wire connections and allowing the electronics to be treated as a single insertable module within the MDF layer stack. Multiple revisions were tested



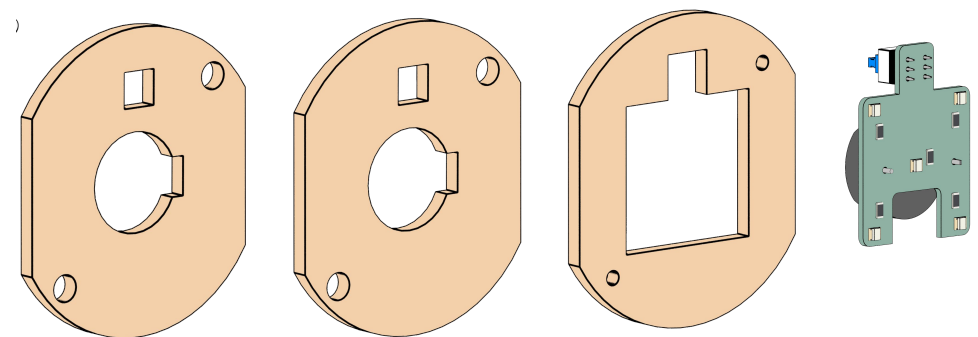
Stage 10 — PCB Dimension Fixing PCB dimensions were iterated across three revisions against the internal cavity of the MDF stack. The board needed to clear cavity walls, align the push switch with its MDF cut-out, and position LEDs at the correct height relative to the diffuser

Stage 11 — Switch and LED Placement Optimisation The push switch was positioned at the top PCB edge, centred on the square MDF cut-out. The five LEDs were spaced to align with the approximate centres of five mandala petal groups — creating an intentional correspondence between glowing petals and LED positions

Stage 12 — Rubber Band Assembly Challenge Early assembly used rubber bands to hold MDF layers during construction. User testing revealed maintaining layer alignment while winding bands required hand coordination inconsistent for children at the lower end of the target age range

Stage 13 — Screw-and-Band Assembly System The method was revised: both M3 screws are inserted through Layer 1 first. Layers 2–4 are stacked directly onto the screw shafts, which act as alignment guides. Rubber-band clips are applied at a defined later stage. This sequencing reduced assembly errors significantly.

Stage 14 — Final Assembly Refinement The final sequence was tested with children aged 8–14 using the production manual. No child required adult assistance. The push-button glow on completion produced an immediate positive response in all participants



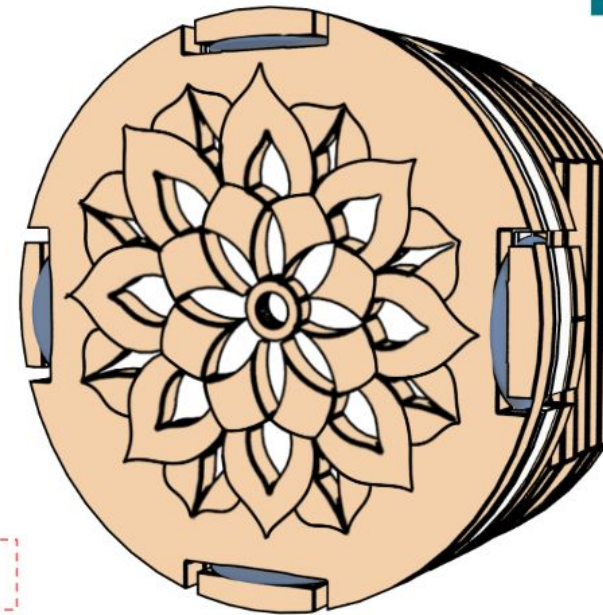
Instruction Manual



Light up Fridge Magnet

DIY Guide

Version 1.1



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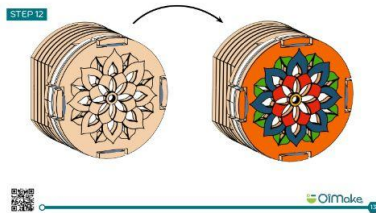
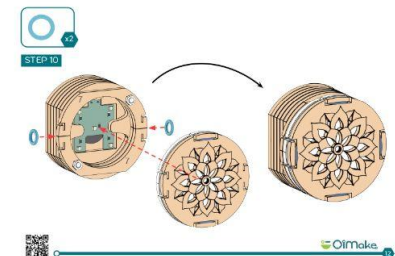
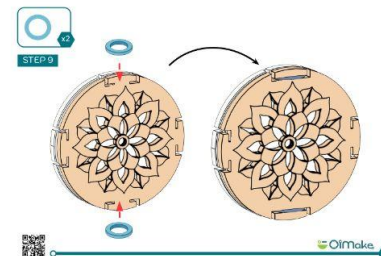
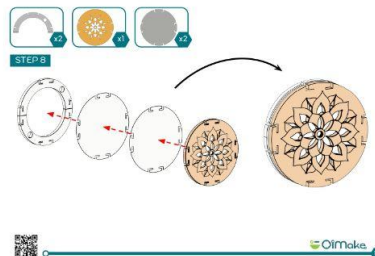
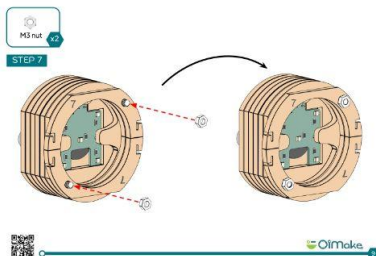
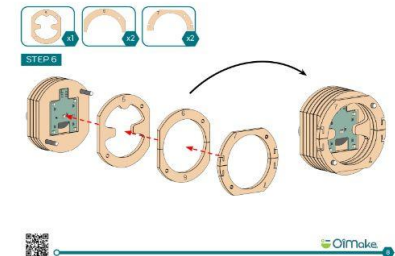
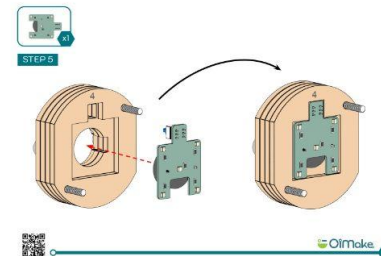
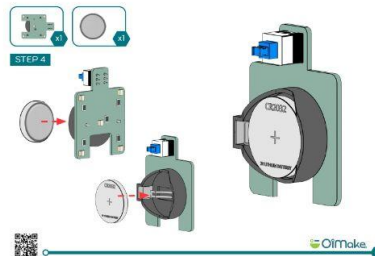
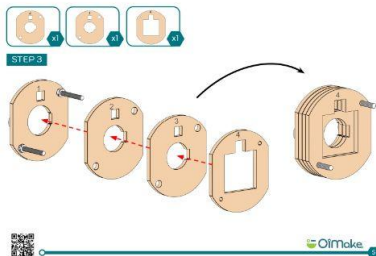
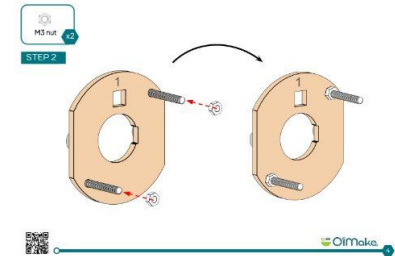
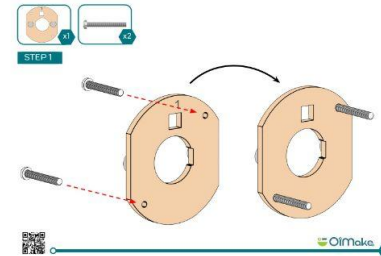
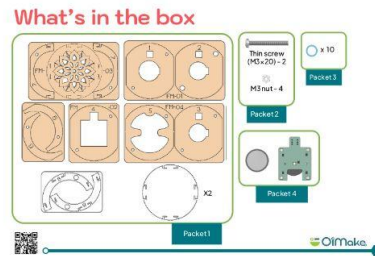
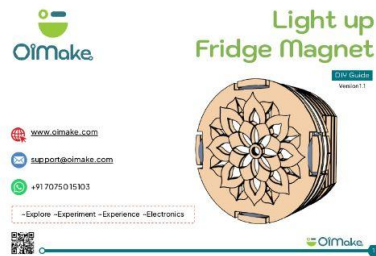
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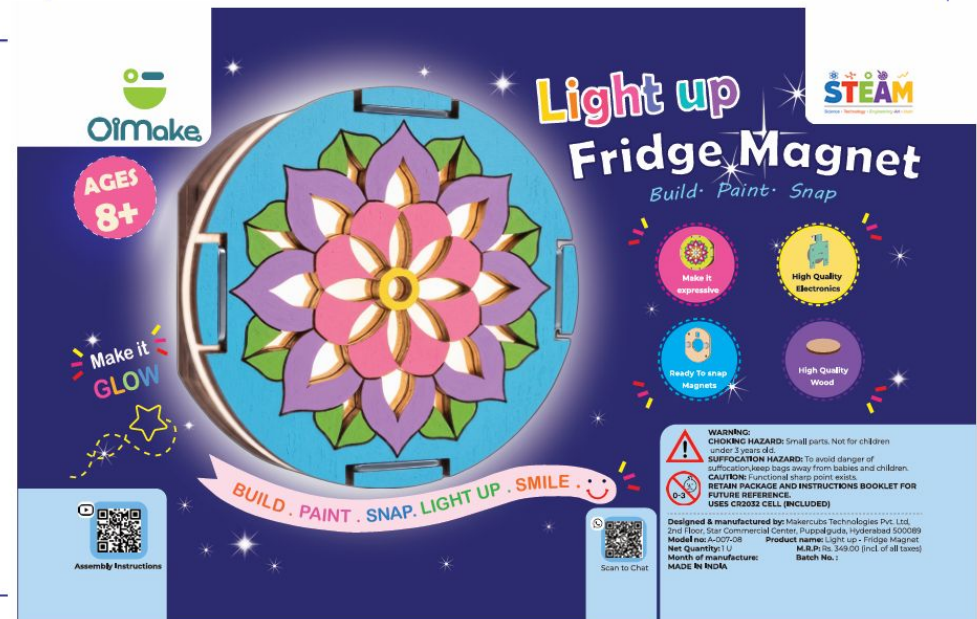
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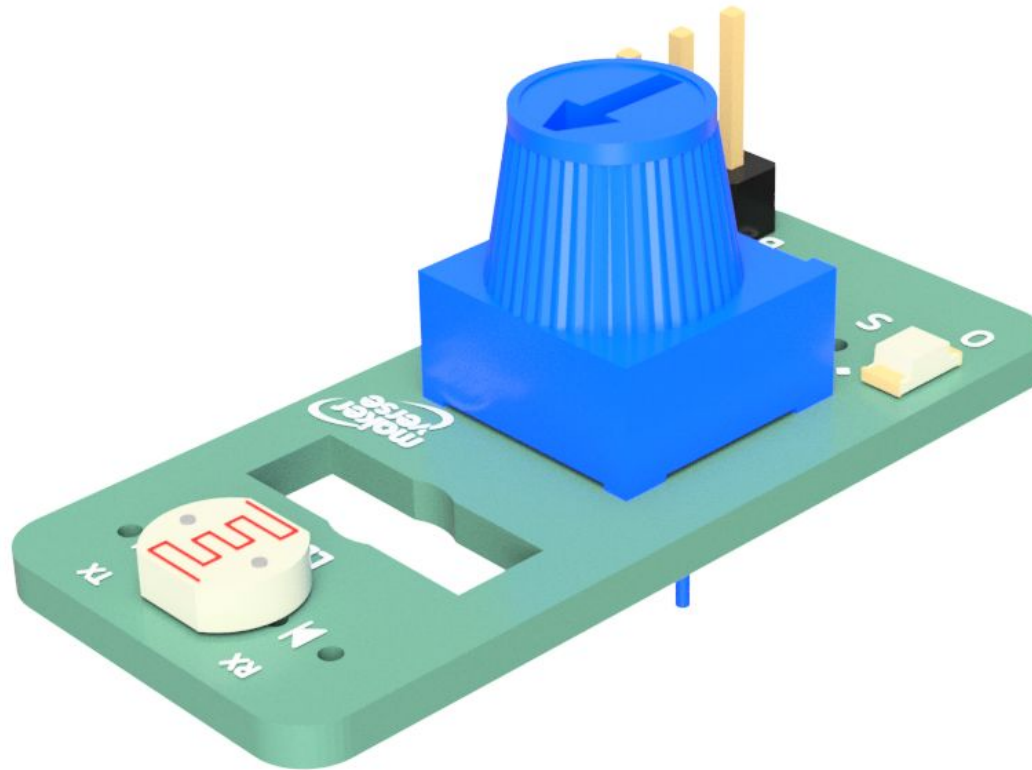


Sticker Design

Instead of sleeve here we made a different approach with sticker over the box , since this a product that falls in the return gift set , we want to make it as low cost as possible, so there is a different idea of providing it in a smaller box.



PROJECT 3



5 in 1 Kit

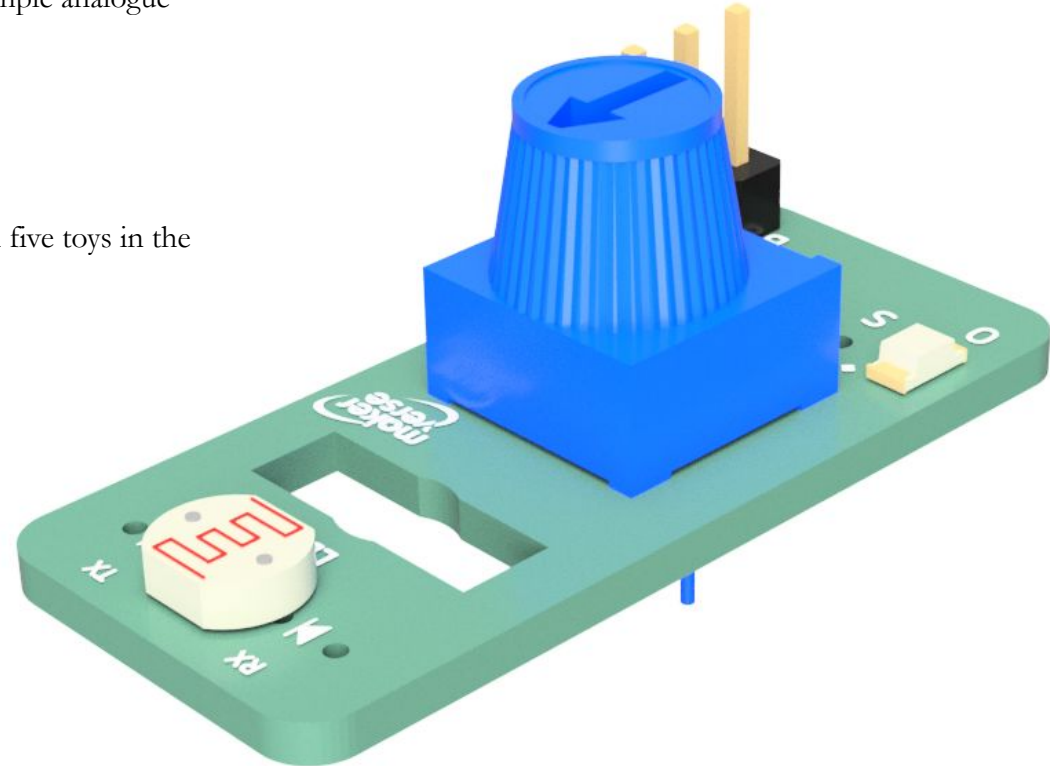
LDR Sensor (Light Dependent Resistor)

The LDR sensor acts as the main input device of the system.

It detects the presence or absence of light and provides a simple analogue output:

- **Light detected → Output ON (1)**
- **No light detected → Output OFF (0)**

This sensor-based interaction becomes the foundation for all five toys in the kit.



Project Overview

The project focuses on developing a **5-in-1 DIY STEAM Toy Kit** that allows children to build multiple interactive toys using a single electronics platform.

The core objective of the project is to introduce children to:

- Basic electronics
- Sensor-based interaction
- Mechanical movement
- Gear mechanisms
- Cause-and-effect systems
- Hands-on DIY assembly learning

The entire kit is designed around a single hero component:

Core Electronics System

Main Control Unit – Motor Driver PCB

The central electronics system of the kit is a custom **motor driver PCB**.

Function of the PCB

The PCB receives the signal from the LDR sensor and processes it to control different output components such as:

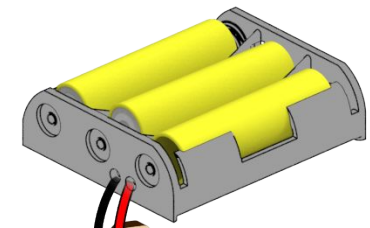
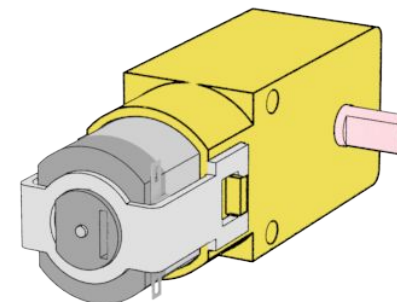
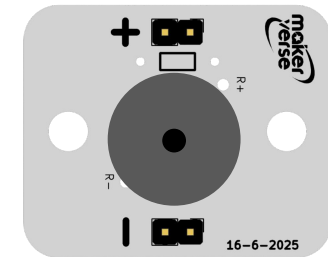
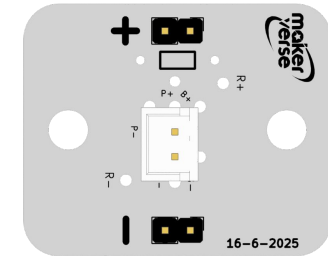
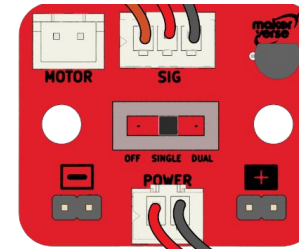
- Motors
- Buzzers
- Rotational mechanisms
- Motion-based actions

The PCB acts as the decision-making and control system of the entire kit.

Working Principle

1. LDR detects light
2. Sensor sends signal to PCB
3. PCB processes input
4. PCB activates assigned output device

Depending on the toy configuration, the output device may vary.



Modular Product System

The project is designed as a **5-in-1 modular kit**.

Instead of creating five separate electronic systems, the same electronics setup is reused across all products.

Important Feature

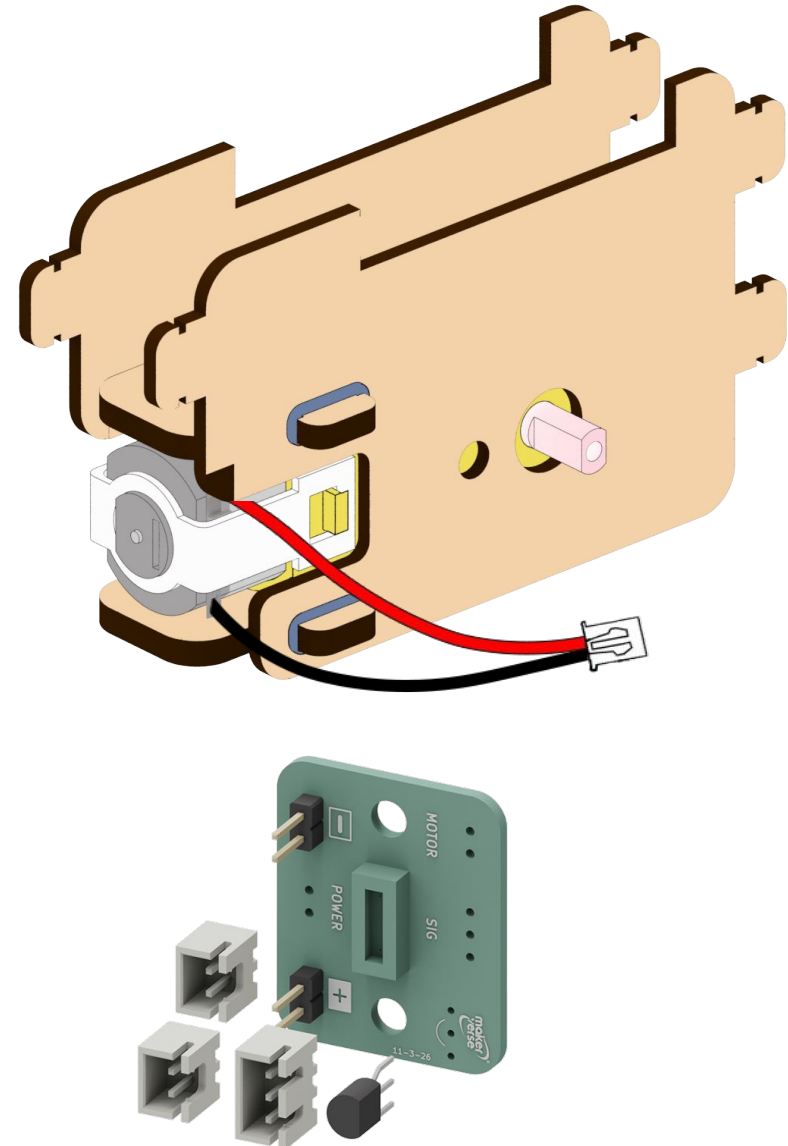
Only **one toy can be assembled at a time**.

If the user wants to build another toy:

1. The current toy must be disassembled
2. Components are reused
3. Another toy configuration can be assembled

This approach encourages:

- Reusability
- Understanding of assembly systems
- Modular thinking
- Sustainable product interaction



Product Distribution

The five products were divided between team members.

Bug Projects (Brief Overview)

The bug toys use the LDR sensor as the primary interaction system.

Working Principle

When light falls on the LDR sensor:

- The PCB receives the signal
- The motor activates
- The bug either:
 - Moves toward the light
 - Moves away from the light

This creates two behavioural responses:

Attraction Behaviour

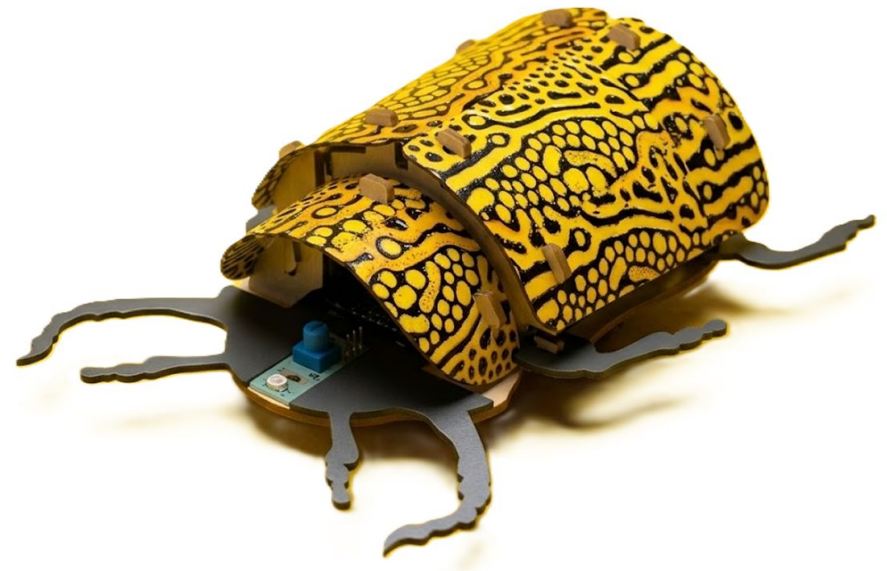
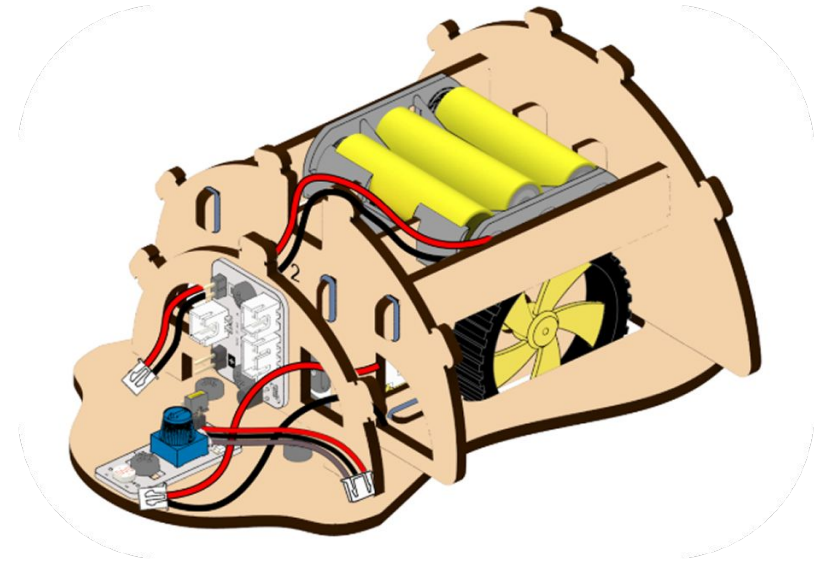
The toy appears attracted to light.

Repulsion Behaviour

The toy appears to escape or repel from light.

These toys help introduce children to:

- Sensor-triggered movement
- Reactive behaviour systems
- Basic robotics concepts



Products Developed in My Contribution

The remaining three products were conceptualized and developed by me:

1. Timer
2. Gramophone
3. Fortune Wheel

These projects explored deeper applications of:

- Timing systems
- Mechanical logic
- Audio interaction
- Uncertainty and randomness
- Gear reduction mechanisms

Timer Project

Concept

The timer project was designed to create a simple one-minute timing mechanism using:

- LDR sensor
- Motor
- Gear reduction system
- Buzzer output

The goal was to make the buzzer activate exactly after one minute.

Initial Challenge

The available motor rotates at:

- **100 RPM**

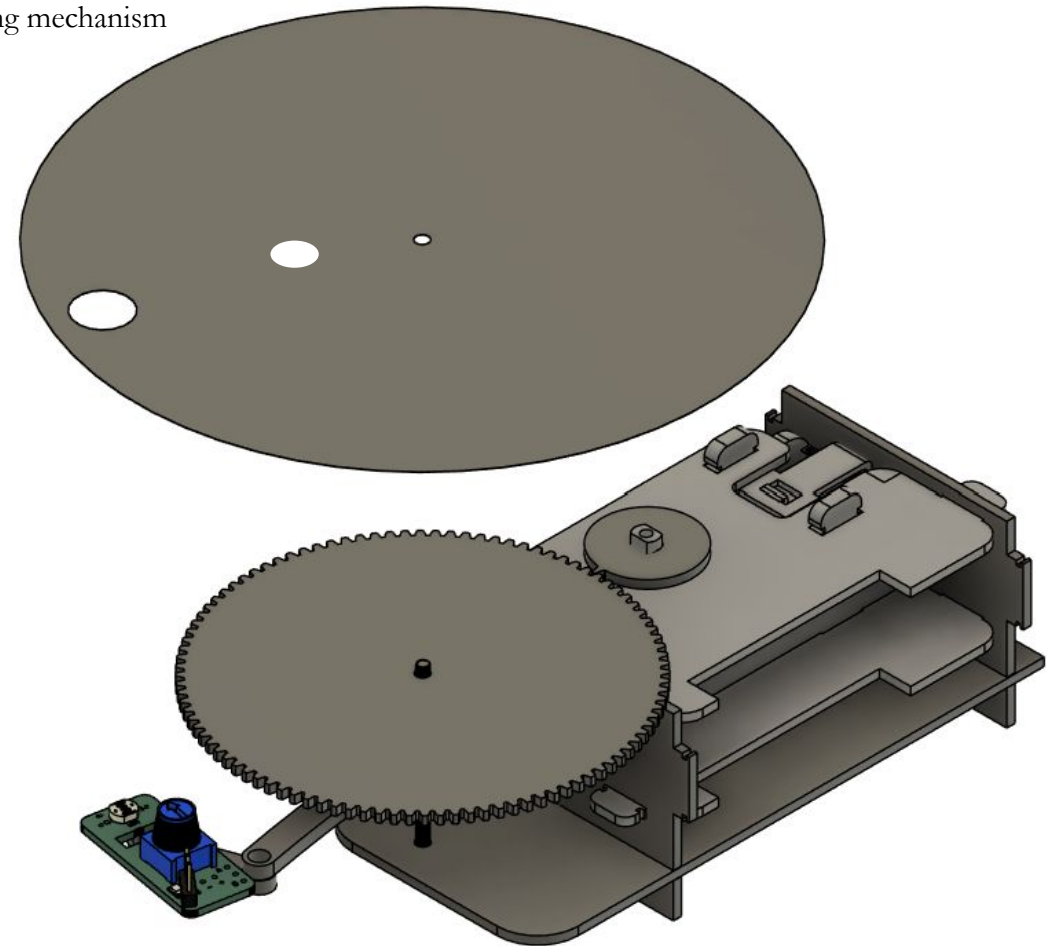
This means the motor completes:

- 100 rotations per minute

However, the project required:

- Only 1 rotation per minute

This created a major speed mismatch.



Gear Reduction Solution

To solve the problem, a gear reduction mechanism was developed.

Gear Setup

- Small driving gear: **1 tooth**
- Large driven gear: **100 teeth**

Working

When the small gear completes 100 rotations:

- The larger gear completes only 1 rotation

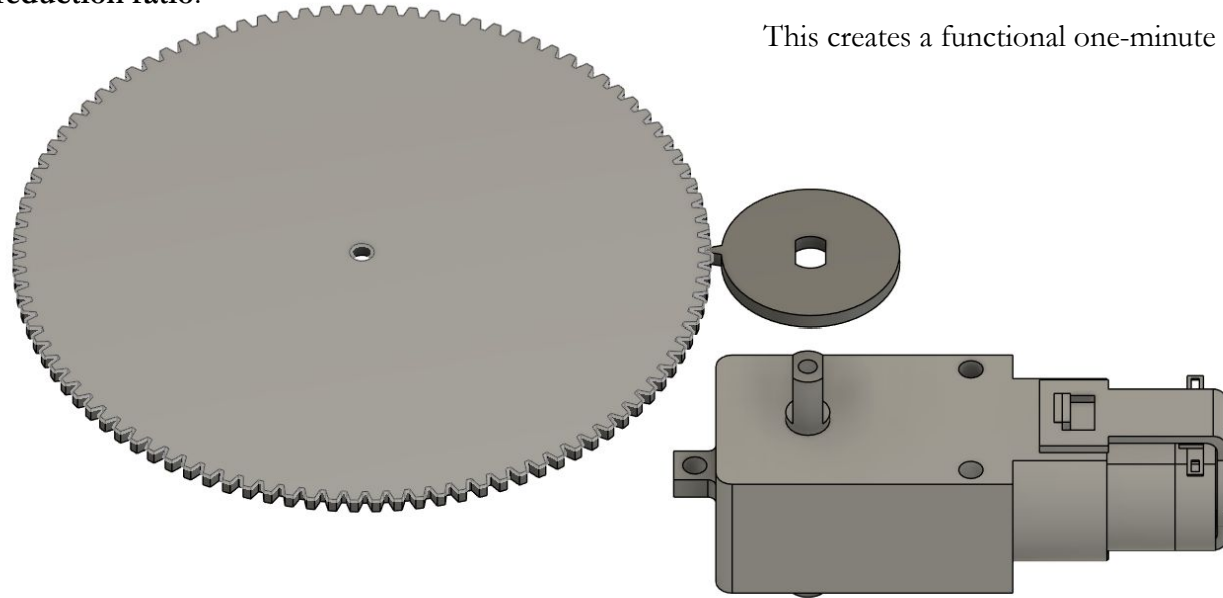
This creates a **100:1 gear reduction ratio**.

Timer Mechanism: A slit opening was created on the rotating disc connected to the larger gear.

Process

1. Motor starts rotating
2. Gear system slows the rotation
3. Large gear completes 1 full rotation in 1 minute
4. Slit aligns with LDR sensor
5. Light reaches the sensor
6. PCB receives signal
7. Buzzer activates

This creates a functional one-minute mechanical timer.



Gramophone Project

Concept

The gramophone project explores the relationship between:

- Rotation
- Light interruption
- Sound generation

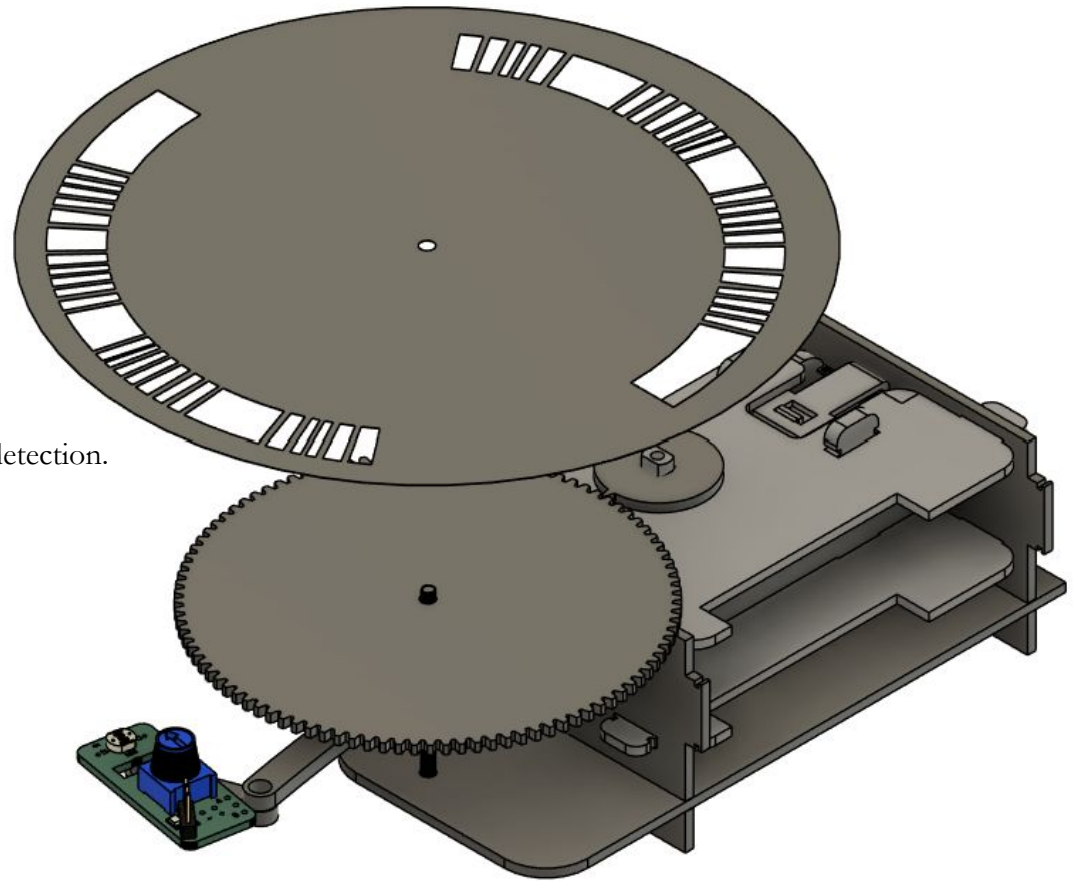
The project simulates a musical system using slits and timed light detection.

System Configuration

Unlike the timer project:

- The motor works independently
- The motor does not depend on LDR input

The LDR sensor only controls the buzzer output.



Working Principle

1. Motor continuously rotates the disc
2. Slits are designed on the rotating disc
3. As slits pass over the LDR:
 - Light reaches the sensor
 - Sensor sends signal to PCB
 - PCB activates buzzer
4. Repeated timed buzzing creates rhythm patterns

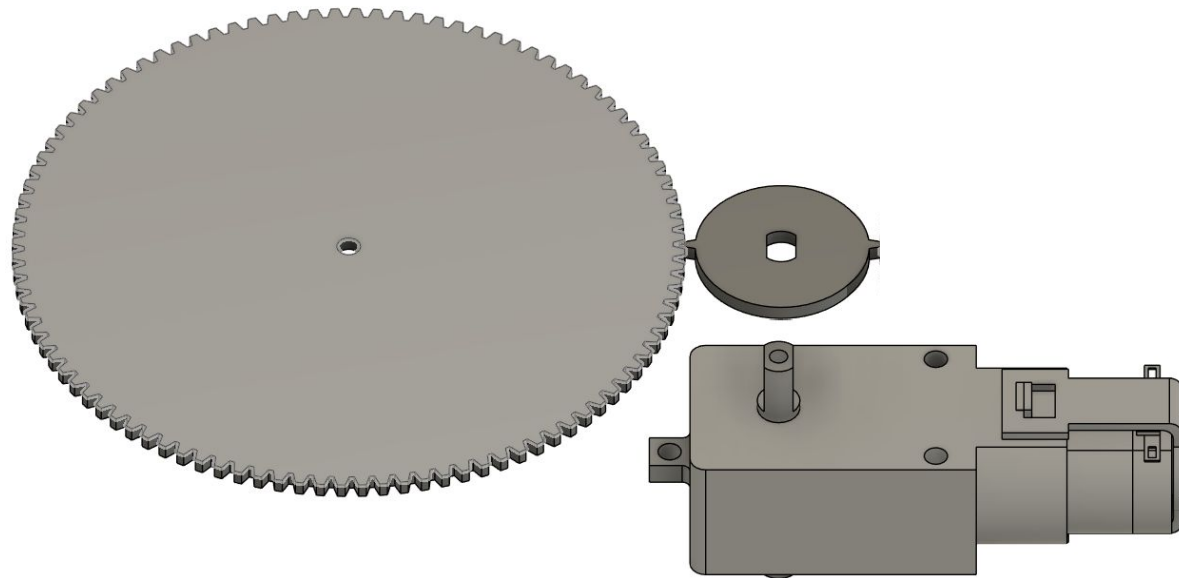
By carefully arranging slit positions and spacing, the buzzer output begins to resemble musical beats or songs.

Gear System Usage

The same gear reduction mechanism used in the timer project was reused here to achieve controlled rotational speed.

This slower and controlled movement allows accurate timing between:

- Slit positions
- Light detection
- Buzzer activation



Fortune Wheel Project

Concept

The fortune wheel project introduces an element of:

- Uncertainty
- Randomness
- Probability-based interaction

Unlike the previous projects, this setup uses manual interaction from the user.

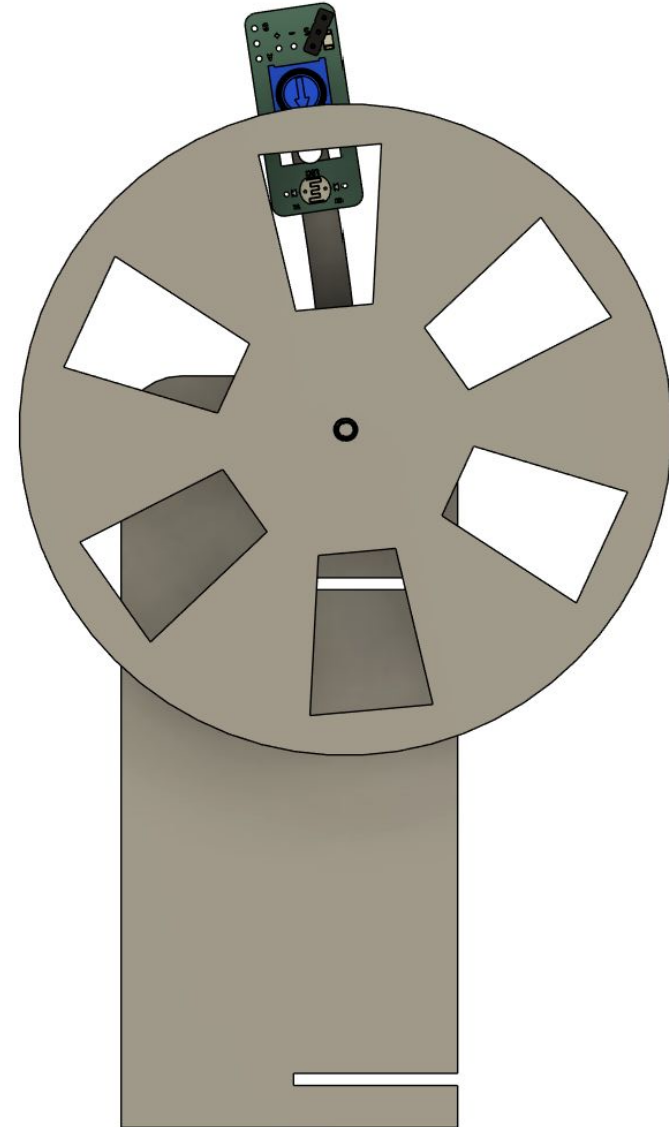
System Structure

The setup contains:

- Circular rotating disc
- Alternating slit pattern
- LDR sensor placed below the disc
- Motor output system

The disc contains repeating sections of:

- Open slit
- Closed surface



Working Principle

User Interaction: The child manually rotates the circular disc.

Sensor Behaviour : As the disc rotates:

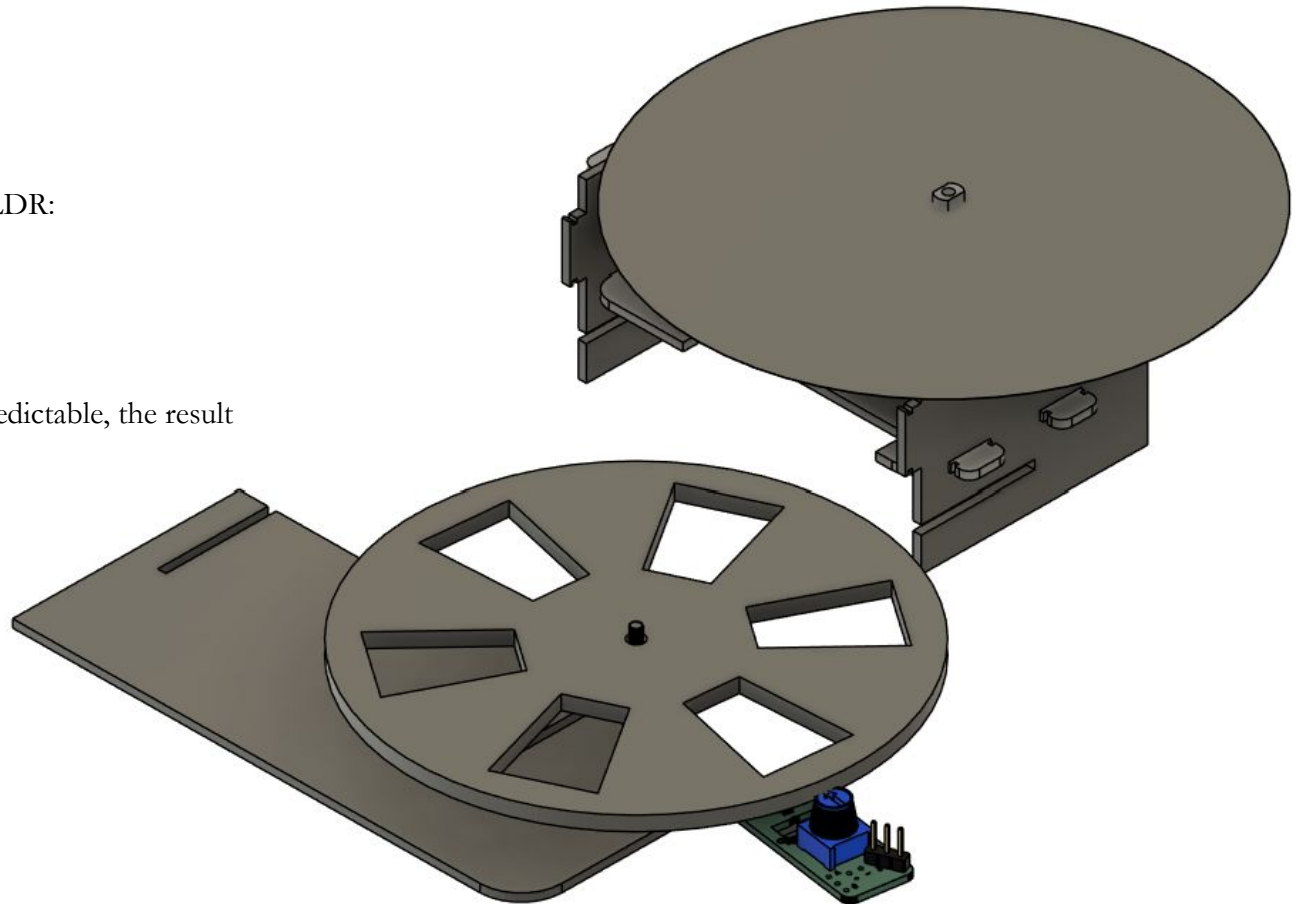
- If the slit aligns above the LDR:
 - Light reaches sensor
 - PCB activates motor
 - Movement occurs
- If the closed section aligns above the LDR:
 - No light reaches sensor
 - No signal is generated
 - Motor remains inactive

Since the stopping position of the disc is unpredictable, the result becomes uncertain.

Importance of Randomness

This project introduces children to:

- Chance-based systems
- Input-output logic
- Manual interaction
- Decision uncertainty
- Interactive gameplay behaviour



Future Scope

The current 5-in-1 DIY STEAM Toy Kit serves as a foundational prototype that demonstrates the integration of electronics, mechanics, and interactive learning through modular toy systems. In the future, the project can be developed into a fully refined commercial educational kit with improved durability, user-friendly assembly systems, and enhanced visual design for children. Additional sensor-based projects can be introduced using the same PCB platform, expanding the kit beyond five products into a larger modular ecosystem. Features such as programmable controls, adjustable speed systems, sound customization, and wireless interaction can also be incorporated to increase educational value. The project also has strong potential for integration into STEAM-based school learning environments, maker spaces, and DIY educational workshops. Future development will focus on improving manufacturability, child safety, packaging experience, and scalability, transforming the concept from a prototype-level learning kit into a complete interactive educational product for wider market adoption.

My Learnings

- **User-Centric Design for Children:** Gained deep insights into the psychology, ergonomics, and behavioral patterns of children. Learned how to design for intuitive interaction, ensuring that complex assembly processes are accessible and engaging for young users.
- **Iterative Prototyping & Technical Integration:** Mastered the transition from low-fidelity mockups to functional prototypes. Navigated the challenges of integrating core electronics (PCBs, LEDs, LDR sensors, and motor drivers) within compact, child-friendly form factors.
- **Design for Manufacturability (DFM) & Cost Optimization:** Learned to balance creative vision with practical constraints through rigorous Bill of Materials (BOM) management, material exploration, and optimizing assembly systems (e.g., the screw-and-band system for the fridge magnet).
- **Visual Communication & Instruction Design:** Developed skills in translating complex mechanical and electrical steps into clear, engaging visual graphics and instruction manuals that empower children to build independently.
- **Real-World Testing & Validation:** Conducted hands-on workshops with kids, learning how to observe user interactions, gather actionable feedback, and seamlessly integrate those insights into final design refinements.

Reflection

The journey through this Graduation Project has been a transformative experience, bridging the gap between academic theory and real-world industrial design. Working on the LED Light Bag, Light-up Fridge Magnet, and the modular 5-in-1 Kit provided a unique opportunity to address a critical modern challenge: the screen time problem.

Watching children interact with the kits during the testing phases was incredibly rewarding. It reinforced my belief in the power of tactile, hands-on learning and the importance of STEAM education. Designing for this demographic taught me that true innovation lies in making the complex feel simple and playful. Navigating the inevitable challenges—from custom PCB dimensional constraints to refining gear reduction systems—taught me resilience and the value of rapid, iterative problem-solving. This project has not only sharpened my technical acumen but also solidified my passion for creating meaningful, precision-driven products that leave a positive impact on the end-user.

Conclusion

Conclusion Design is not just about creating products; it is about crafting experiences that educate, inspire, and empower. This project stands as a testament to the potential of design to foster curiosity and hands-on making in the next generation.

Thank You!

A sincere thank you to everyone who supported this journey:

- **OiMake** for the opportunity, resources, and collaborative environment.
- **Prof. Bhushan Sonawane** for the continuous guidance, mentorship, and support.
- **Woxsen University (School of Arts and Design)** for providing the foundational knowledge and platform to grow as a designer.
- **The Jury and Audience** for your time and valuable feedback.

